Highlights

Anchored Conditionally-Gated Control for Wheeled Bipedal Balance and Disturbance Recovery

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- Gated damping delivers the steadiness while the anchor integral delivers accuracy.
- A chiral push envelope is traced to a sign convention and closed by mirror controls.
- Closed-form model predicts the 27 mm standing offset to 0.92% over a 40x sweep.
- 0.294 mm standing repeatability, held as a fixed point for 30 min without falling.
- Recovers 108 N median pushes, a 100 cm free fall, and 10-20 cm curbs per-leg.

Anchored Conditionally-Gated Control for Wheeled Bipedal Balance and Disturbance Recovery

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ABSTRACT

Wheeled bipedal robots confront a design trade-off in which the stiffness that buys precise standing shrinks the push-recovery envelope, while the integral action that removes equilibrium bias winds up during disturbances. The Anchored Conditionally-Gated Controller answers both through conditional activation. Its position integral is confined to a 15 cm neighbourhood and enabled by an asymmetric envelope follower whose 23 ms attack and 1.5 s release separate quiet stance from post-push ringdown. On a 10-DOF, 8.1 kg wheeled biped in MuJoCo the architecture is dissected rather than merely demonstrated. A factorial ablation separates three coupled effects. Gated damping produces the steadiness, the anchor integral produces the accuracy at a threefold cost in steadiness, and the gates are bit-for-bit inert in quiet stance yet decisive under disturbance. The push envelope is measurably chiral at −17.3 N, traced to the antisymmetric hip-roll and hip-yaw sign convention and closed by two controls, since a mirror flips the asymmetry and a sham mirror does not. A closed-form model predicts the 27.0 mm standing offset to 0.92 % across a fortyfold gain sweep and identifies it as the price of a bounded integrator memory. ACC holds 0.294±0.015 mm sagittal repeatability over ten trials, a 59-fold improvement on the proportional-only rung of the same ablation, recovers omnidirectional pushes at 108 N median and 82 N worst case, stands 30 min at 0.0004 mm/min drift where six classical baselines and a full-state LQR all fall, and survives a 100 cm drop as well as curbs of 10 – 20 cm. A 2 ms-resolved delay screen prices deployment at a latency cliff of 6 – 8 ms that a single retuned gain moves outward, making the cliff a tuning artefact rather than an architectural ceiling.

CRediT authorship contribution statement

Van Thuong Nguyen: Conceptualization, Methodology, Software, Validation, Investigation, Data curation, Writing – original draft, Visualization. **Nhu Thanh Vo:** Conceptualization, Methodology, Validation, Formal analysis, Investigation, Data curation, Writing – review and editing, Visualization.

1. Introduction

Wheeled biped robots are expected to stand with millimetre precision, to survive pushes from any direction, and to adapt to uneven ground, yet no single classical controller delivers all three at once. LQR is the usual answer for balance [1], whole-body control for posture [2] and MPC for terrain [3], while locomotion is reached for separately again through learned policies [4].

The obstacle is structural rather than a matter of tuning. Lowering the proportional gain widens the capture basin at the cost of balance stiffness, and omitting integral action leaves a 1.3 Nm equilibrium bias that sustains a persistent limit cycle whatever that gain is set to, a cycle that is predominantly sagittal at 69.7 mm peak to peak along the rolling direction against 13.4 mm laterally. It lives on the one axis the wheels can drive, which is both why it persists and why every mechanism developed here is aimed at it, and restoring a global integral removes the bias but winds up during a push, so the two obvious remedies exclude one another.

This paper presents the Anchored Conditionally-Gated Controller, which resolves both dilemmas through conditional activation. Its central mechanism is a proximity-gated anchor, a position integral confined to a 15 cm neighbourhood of the commanded home and enabled by an asymmetric envelope follower whose 23 ms attack and 1.5 s release distinguish quiet stance from post-push ringdown. Coupled with a gated damping boost, ACC reaches 0.294±0.015 mm repeatability on the sagittal axis under clean sensing and zero actuator delay over ten trials, at push performance equal to or better than the proportional-only rung that carries no integral action at all. Absolute accuracy on that axis is a separate quantity. The robot parks 27.0 mm from the commanded home and then holds that offset to 0.009 mm, a behaviour measured in Section 5.1 and derived in closed form in Appendix C.

A factorial ablation run under a single protocol, reported in Table 2, locates the mechanism more precisely than the gate structure alone would suggest. Quiet-stance steadiness comes from the gated damping terms rather than from the anchor integral, since disabling the integral while leaving every gate intact improves sagittal repeatability from 0.294 mm to 0.089 mm whereas reverting the two velocity-damping coefficients reinstates the full 25.2 mm limit cycle. What the integral buys is accuracy, removing 4.2 mm of sagittal parking offset at a threefold cost in steadiness, and because the two properties come from different mechanisms and trade against each other they are reported as separate columns rather than as a single planar number. The proximity gate

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and the asymmetric envelope are by contrast inert during quiet stance, ablating either reproducing ACC bit for bit because the robot never leaves the gate's inner band nor the envelope's release branch while standing still, so their contribution is windup prevention and ringdown control that only a displacement experiment can measure. The pitch safety gate is load-bearing even with no disturbance applied and removing it produces divergence from initial-condition noise alone in all ten trials. The gates are therefore enabling conditions that make an otherwise inadmissible damping and integral configuration safe, rather than the proximate source of the precision.

Against that architecture we place six classical baselines, four derived from a 4-state TWIP model and two from a 6-state coupled model, all of which fall in every trial, removing the PID servo path in the direct-torque variant buying only 0.28 s of survival at an unchanged fall rate. A seventh design, a full-state LQR linearized from the plant itself, removes the model-mismatch objection in Section 5.4.2 and survives two orders of magnitude longer while still falling, so platform complexity rather than model order or servo lag dominates the failure.

The work is positioned within structured classical priors rather than within learned control, its object of study being a hand-designed and fully interpretable torque assembly whose gating encodes the plant's measured failure modes explicitly. It was built to serve as the nominal base action of a residual-learning stack, where a policy corrects a structured prior instead of searching the space unaided [8, 9], and that origin explains its shape, since such a prior wants to be interpretable, bounded and gated rather than optimal. What the paper reports is the prior itself, no residual policy being trained here, so the classical controllers are the comparison of record. A preliminary PPO [5] run of 5.4M steps on a single seed motivates the structure without bounding it, standing at one height for about 3.20 s against 0.16 – 0.60 s for the classical baselines but failing to generalize across the 0.40 – 0.70 m base-z range of Section 3.1, with height RMSE above 50 mm and falls on 85 % of squat transitions. Those failures concentrate on exactly the coupled pitch and height dynamics that a posture map and a gate schedule encode explicitly, and since the run is single-seed and carries neither domain randomization [6, 7] nor a hyperparameter search it is retained as a difficulty probe rather than as a bound.

The contributions are threefold. First, a proximity-gated anchor with an asymmetric envelope follower attains sub-millimetre idle precision without sacrificing omnidirectional push survival. Second, a two-channel conditionally-gated torque assembly carries flight recovery and per-leg terrain adaptation as independent extensions, each ablated to establish which mechanism is necessary for which behaviour, with thirty repetitions on push and ten on idle and terrain. Third, a factorial ablation combining an additive ladder L0–L3 with a subtractive ladder S0–S5 is run against the six classical baselines and supplemented by two control experiments that test the comparison itself, namely a full-state LQR derived from the plant rather than from a reduced model and a sweep of the coupled model's single empirical constant.

The remainder of the paper is organized as follows. Section 2 reviews wheeled bipedal control and the disturbance-rejection literature ACC departs from, Section 3 describes the robot and the simulation setup and Section 4 develops the controller. Section 5 reports the experimental validation, from anchored standing through push, flight, terrain, baseline comparison and architectural analysis, and Section 6 states the limitations and the deployment consequences that follow from them, with three appendices supplying the anti-windup argument, the whole-body control baseline details and the closed-form account of the standing offset.

2. Related Work

2.1. Wheeled Bipedal Platforms and Classical Control

Ascento [1] demonstrated two-wheeled jumping under LQR balancing and was later extended to whole-body control with kinematic loops [2]. Handle [10] pioneered wheeled-leg coordination, and wheeled quadrupeds combined driving with walking through whole-body motion control and planning [11]. Ollie [12] adapts whole-body balance control to a two-wheeled biped, while wheeled quadrupeds have since been driven by whole-body MPC with online gait sequencing [13]. DIABLO [14] pairs an LQR balance controller with separate yaw, split-angle, height and roll loops, and SKATER [3] uses a hierarchical MPC whole-body controller with centrifugal-force compensation and terrain adaptation. Closest to our morphology is Whleaper [15], which also carries ten degrees of freedom with three hip axes per leg. It applies LQR to balanced sliding and a separately trained PPO policy to walking, switching between the two by locomotion mode rather than composing them on a shared control output. Composing a classical prior with a learned residual on one output is instead the residual-RL construction [8, 9], applied recently to wheeled-biped balance [16], and ACC is designed to supply the classical prior in that construction rather than to replace the learned part. Placing ACC beside these platforms on the capabilities measured here is possible only in a limited sense, because none of the cited works reports idle standing precision, a quantitative push threshold or a per-leg terrain estimate, their balance layers being LQR on Ascento, LQR with auxiliary loops on DIABLO, mode-switched LQR and PPO on Whleaper and hierarchical MPC with vision-based terrain adaptation on SKATER. The one comparison that runs cleanly in the other direction is validation, since all four are demonstrated on hardware whereas every result here is in simulation.

Coupled pitch–roll LQR is standard on rigid two-wheeled platforms, where torque acts directly on the wheels and there are neither articulated legs nor leg-height dynamics, which makes for a structurally simpler problem with fewer unstable modes. Our 6-state LQR reproduces that coupled structure on a legged morphology and fails, and the 4-state direct-torque variant, which removes the servo path and re-optimizes the roll gain to $k_{p,\mathrm{roll}}$=55.0, fails as well at a full fall rate and 0.60 s of survival. Since that variant carries no servo lag whatsoever and still falls, as Section 5.4 shows, actuation lag alone cannot account for the outcome and the articulated-leg morphology with its height dynamics is the dominant remaining factor. Both variants take their gains from a simplified 4-state TWIP model rather than from a full 10-DOF linearization, an objection removed by the control experiment of Section 5.4.2.

Cascade control [17] and whole-body control through hierarchical inverse dynamics or task-priority quadratic programs [18–20] use fixed-gain nested or strictly prioritized loops without conditional gates, while gain scheduling [21, 22] varies parameters continuously with one scheduling variable rather than responding to qualitatively distinct physical regimes. We benchmarked a task-priority QP whole-body controller both standalone and as an assist layer on ACC in Section 5.5, where it fails to resolve the precision and recovery trade-off because its fixed formulation carries no conditional regime gating. Split-range, selector and override architectures [23, 24] already implement continuous blending in process control and ACC adopts that principle, its contribution being the synthesis of physically motivated gating surfaces, one spatial in proximity, one temporal in the velocity envelope and one a pitch safety interlock, derived from wheeled-biped failure modes rather than from generic process-variable limits. The gates also connect ACC to hybrid and switched systems [25] and to hysteresis-based switching logic [26], which deliberately breaks the symmetry of a switching surface to prevent chattering, an asymmetry ACC realizes in continuous time through its fast-attack and slow-release envelope with smoothstep softening the surfaces to C^1 transitions. Section 5.7 makes that connection quantitative by locating the operating point relative to every switching surface and settling the one switch that does fire with a common Lyapunov function. A recent review [27] surveys wheel-legged biped platforms and notes the absence of a unified classical solution.

2.2. Disturbance Rejection, Anti-Windup, and ACC's Distinction

Disturbance observers [28, 29] and active disturbance rejection [30, 31] provide temporal disturbance estimation but cannot distinguish a push from the ringdown that follows it, because both occupy the same frequency band. Classical anti-windup in its back-calculation, conditional-integration and unified observer-based forms [32–34] triggers on actuator saturation, whereas ACC's anchor gates spatially and on physics. The proximity gate confines integration to a 15 cm neighbourhood and the asymmetric envelope separates quiet stance from ringdown by velocity envelope, which replaces the estimation-speed against noise trade-off with a time-domain attack against release trade-off. Push recovery through capture point [35, 36] and flight-phase control [37] is complementary, and blind stair climbing by reinforcement learning has been shown on Ascento [38]. Terrain adaptation is usually built on exteroceptive elevation mapping [39, 40], on learned policies that fuse proprioception with terrain estimates [41, 42], or on purely proprioceptive contact sensing [43–45], and ACC extends proprioceptive contact-point estimation to wheeled bipeds. Hyon and Cheng [46] achieved full-body humanoid balancing by gravity compensation with optimal contact-force distribution, making the robot passive to external forces without measuring them. ACC shares the requirement of reacting to unmeasured pushes but meets it by switching regime on proprioceptive gating variables rather than by redistributing contact forces.

3. Robot Platform

3.1. Robot Morphology and Simulation

The robot shown in Fig. 1 is a ten-DOF wheeled biped carrying five actuated joints per leg, namely hip roll, hip yaw, hip pitch, knee and drive wheel. It has a mass of 8.1 kg, a 0.26 m thigh, a 0.28 m shin, wheels of 0.06 m radius and a hip width of 0.23 m. Torque is limited to 30 Nm on the hip-roll, hip-yaw and wheel axes and to 150 Nm on hip pitch and knee.

Simulation runs in MuJoCo [47] at 500 Hz. ACC and every classical baseline are controlled at 100 Hz over five physics substeps, the classical baselines are additionally measured at their 50 Hz design rate in Section 5.4.1, and the PPO reference is reported at the 50 Hz rate it was trained at, with a 400 Nm/s torque rate limit and raw torque commands throughout. The integrator is implicitfast with four solver and eight linesearch iterations, the friction cone is pyramidal for the body and elliptic for the wheels with μ_s=1.2 and μ_k=0.8, armature is 0.008 kg·m^2 on the wheels and 0.02 kg·m^2 on the leg joints, and contact stiffness follows `solref`={0.002,1} on the body and {0.005,0.7} on the wheels. That stiff body contact idealizes rigid ground, and softening `solref[0]` to 0.02, which approximates tire on asphalt, raises planar idle RMS by roughly 60 % to about 1.2 mm, so every sub-millimetre idle figure reported here is conditional on stiff ground contact.

Height convention

Two vertical references appear in this paper and are easily confused, so we fix them here and mark every height accordingly. CoM-z is the height of the whole-body centre of mass above ground and is the quantity ACC commands and regulates, with the nominal standing pose at h_{CoM}=0.404 m, whereas base-z is the torso root-body height `qpos[2]` in the MuJoCo model, at which the same pose sits at z_{base}=0.536 m, 13.2 cm above the centre of mass. The offset is pose-dependent but varies by less than 1 cm over the postures used here. All ACC results are reported in CoM-z, whereas the PPO reference of Section 5.4 commands base-z over 0.40 – 0.70 m and the WBC baselines of Section 5.5 inherit that convention, so those numbers are labelled base-z where they appear and must not be compared numerically against CoM-z heights, substituting one for the other in

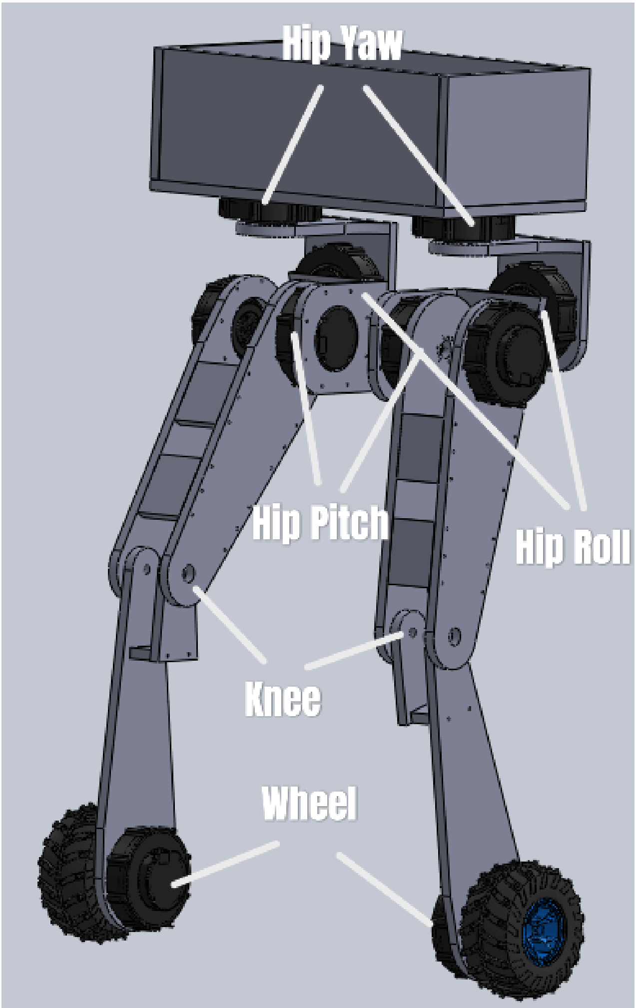

Figure 1: Robot joint layout with world coordinate frame (right-handed, Z up). In the model as built, both wheel axles lie along world X and the two wheel links are separated by 0.271 m along X; the rolling (and therefore sagittal) direction is world Y, and world X is the lateral (track) axis. All axis-resolved results in this paper follow that convention, verified four independent ways in Section 5.1. The CoM is located at the torso geometric center. Each leg: hip roll (HR), hip yaw (HY), hip pitch (HP), knee (KN), wheel (WH).

a height command injecting the full 13.2 cm offset as a reference error.

ACC's posture map is calibrated at two setups, at CoM heights of 0.354 m and 0.454 m (±5 cm about nominal), and interpolated between them. Outside that band the joint targets are linearly extrapolated, bounded to the interval [0.254, 0.554] m, i.e. 10 cm beyond the calibrated band at each end. This ±10 cm extrapolation bound, and not a ±5 cm one, is what the per-leg height split (Section 4.6) operates against.

4. ACC Formulation

4.1. Two-Channel Torque Assembly

ACC assembles a PD balance core, a proximity-gated anchor integral and posture regulation into two torque channels, the wheel torque $\boldsymbol{\tau}_w \in \mathbb{R}^2$ and the leg-joint torque $\boldsymbol{\tau}_q \in \mathbb{R}^8$, with flight and terrain entering as independent extensions as sketched in Fig. 2,

$$\begin{aligned}\boldsymbol{\tau}_w &= (1-g_{\text{flight}})\big[\boldsymbol{\tau}_{\text{balance}} + g_{\text{anchor}}\boldsymbol{\tau}_{\text{anchor}}\big] + g_{\text{flight}}\boldsymbol{\tau}_{\text{flight}},\\ \boldsymbol{\tau}_q &= \boldsymbol{\tau}_{\text{posture}}\big(h^{\text{cmd},L},\, h^{\text{cmd},R}\big) + \boldsymbol{\tau}_{\text{lat}} + \boldsymbol{\tau}_{\text{yaw}}.\end{aligned} \tag{1}$$

Two structural points are easy to misread as sums. The flight term is an override rather than a summand, so at g_{flight}=1 it replaces the wheel torque instead of adding to it, a deliberate replacement since the balance core regulates sagittal position and velocity against a ground that unloaded wheels do not have, while the convex blend makes the substitution continuous. Per-leg ground adaptation likewise enters as a reference split rather than as an extra torque, the posture regulator always taking two height commands that coincide on level ground and separate through Eq. (22) across a step, while the gate g_{terrain} appears not in $\boldsymbol{\tau}_q$ but in the wheel channel, where it retunes the sagittal loop for the straddle treated in Section 4.6.

Every gate g_* except the flight gate is a smoothstep function of a physical condition such as proximity, quietness, attitude or terrain disparity rather than a discrete switch,

$$\text{ss}(x;a,b) = 3t^2 - 2t^3, \qquad t = \text{clip}\Big(\frac{x-a}{b-a},\, 0,\, 1\Big). \tag{2}$$

Equation (2) is the standard Hermite blend, C^1 in x, equal to zero for $x \le a$ and to one for $x \ge b$, with vanishing derivative at both knees. Continuity matters for a concrete reason, because a hard threshold on any of these conditions injects a torque step whose magnitude equals the full gated component, and at 100 Hz under a 400 Nm/s rate limit the rate limiter spreads that step over several control cycles and delivers it to the plant as an unmodelled impulse. Smoothstep removes the step entirely and the rate limiter is therefore never the active constraint during a gate transition. The one exception is g_{flight}, and only on the way in, since contact loss is a binary physical event, so engagement is a latched load-based hysteresis that arms after two consecutive steps with both wheels unloaded and releases after five steps at $F_z \ge 0.5mg$, an asymmetry that exists specifically to prevent balance torque from re-engaging during landing bounce. The handback is not discrete either, g_{flight} ramping linearly to zero over fifteen steps after release, because an instant handback at large pitch would return roughly 19 Nm of wheel authority to a marginally loaded wheel and relaunch the robot. One caveat applies to the composition $g_{\text{env}} = 1 - \text{ss}(\text{EMA}(|v_{\text{sag}}|);\cdot)$, which is only C^0 in time because the asymmetric EMA of Eq. (15) switches coefficients when $|v_t|$ crosses EMA_{t-1}. The smoothstep remains C^1 in its argument, so the resulting torque is continuous while its time derivative may jump at an attack or release transition.

Wheel torques control sagittal balance, whereas lateral and yaw regulation sense torso state but actuate in the leg-joint channel through the antisymmetric hip-roll and hip-yaw pairs of Eq. (7), so the wheels carry no roll or yaw differential. On level ground the two channels are coupled only through the plant, since with $h^{\mathrm{cmd},L}{=}h^{\mathrm{cmd},R}$ the terrain gate is identically zero and no term in $\boldsymbol{\tau}_w$ reads a leg-posture state. Across a step the gate becomes a function of $|h^{\mathrm{cmd},L} - h^{\mathrm{cmd},R}|$ and reaches into the wheel channel in four places, softening k_{pos}, raising the position-torque cap, adding a pitch-rate feedforward to the wheel-velocity setpoint and symmetrising the per-wheel damping so that a one-wheel step is not read as a yaw command, a coupling confined to the straddle and vanishing continuously with the height split, which is what allows the extensions of Sections 4.5 and 4.6 to be added without retuning the anchor.

Conditional gating here denotes smoothstep activation by physical conditions, namely g_{prox}, g_{env}, g_θ and g_{terrain}, and it differs from the industrial override architectures of Section 2.1 in gating components within a single torque assembly by physical regime rather than selecting among parallel controllers by process-variable limits. The gating variables are not control errors but physical quantities of three different kinds, one spatial, one temporal and one a safety interlock on attitude, each chosen because a specific measured failure mode of this platform is expressible in it.

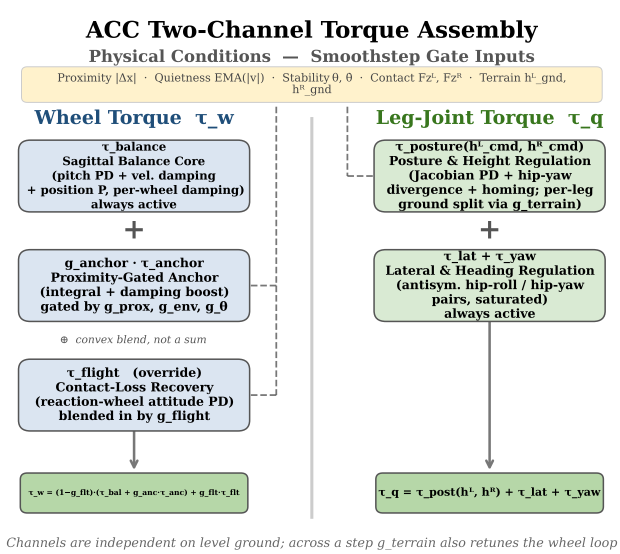

Figure 2: ACC two-channel torque assembly. Physical conditions gate each component via continuous smoothstep functions (dashed). The flight gate is the one latched exception, and it *overrides* the wheel channel rather than adding to it; per-leg ground adaptation enters the posture regulator as a split pair of height references, not as an added torque.

4.2. Sagittal Balance Core (τ_{balance})

The sagittal balance component is always active and acts entirely in the wheel-torque channel,

$$\boldsymbol{\tau}_{\mathrm{balance}} = \begin{bmatrix} \tau_{\mathrm{com}} + \tau_w^L \\ \tau_{\mathrm{com}} + \tau_w^R \end{bmatrix}, \qquad \begin{aligned} \tau_{\mathrm{com}} &= \tau_{\mathrm{pitch}} + \tau_{\mathrm{damping}} + \tau_{\mathrm{pos}}, \\ \tau_w^i &= -k_w\big(\omega^i - \omega^{\mathrm{cmd}}\big). \end{aligned} \tag{3}$$

where τ_{com} is the common-mode balance torque, identical on both wheels, and τ_w^i for $i \in \{L, R\}$ is the per-wheel velocity damping toward the commanded wheel speed, the only term that differs between the two wheels. The individual terms are

$$\tau_{\mathrm{pitch}} = k_p \cdot \theta + k_d \cdot \dot{\theta}, \tag{4}$$

$$\tau_{\mathrm{damping}} = -\big(s_v k_{\mathrm{vel}} + k_{v2}\big) \cdot v_{\mathrm{sag}}, \tag{5}$$

$$\tau_{\mathrm{pos}} = -k_{\mathrm{pos}} \cdot \mathrm{clip}(\Delta x,\ \pm 0.10\,\mathrm{m}), \tag{6}$$

where θ is the torso pitch angle, $\dot{\theta}$ the pitch rate, v_{sag} the sagittal body velocity and Δx the sagittal position error from the latched home position. Two coefficients damp v_{sag}, the scheduled term $s_v k_{\mathrm{vel}}$ whose scale s_v the M2 and M3 ablations of Section 5.1.5 revert, and a fixed offset k_{v2} that is never scheduled or gated.

Lateral and yaw regulation read torso state alongside the balance core but actuate in the leg-joint channel, antisymmetrically across the hip-roll pair $(\mathrm{HR}^L, \mathrm{HR}^R)$ and the hip-yaw pair $(\mathrm{HY}^L, \mathrm{HY}^R)$,

$$\begin{aligned} \boldsymbol{\tau}_{\mathrm{lat}} &= \begin{bmatrix} +1 \\ -1 \end{bmatrix} \big(k_{p,\mathrm{roll}}\phi + k_{d,\mathrm{roll}}\dot{\phi}\big), \\ \boldsymbol{\tau}_{\mathrm{yaw}} &= \begin{bmatrix} -1 \\ +1 \end{bmatrix} \big(k_{p,\mathrm{yaw}} e_\psi - k_{d,\mathrm{yaw}}\omega_z\big), \end{aligned} \tag{7}$$

each saturated at its own moment limit, with ϕ the roll angle, e_ψ the yaw error and ω_z the yaw rate. On this morphology the lateral axis is closed by the legs rather than by a wheel differential. It is the axis on which six classical baselines fail in Section 5.4, and the fixed sign pattern of these two antisymmetric pairs is the mechanism behind the chirality of the push envelope reported in Section 5.2.

The proportional position term, clipped to ± 0.10 m and saturating at ± 4 Nm, keeps the robot within 4 – 6 cm of home but leaves a 1.3 Nm equilibrium bias that sustains a perpetual limit cycle, which is what motivates the anchor integral. Classical anti-windup does not address it, because such schemes fire on actuator saturation whereas the critical failure on this platform is state-dependent, the position error growing to 10 – 30 cm during a push while the wheels stay well inside their 20 rad/s range, so a saturation-triggered scheme never activates at all. ACC's proximity gate supplies spatial confinement through a continuous smoothstep instead, avoiding both the over-restriction that would block slow-drift correction and the under-restriction that would allow windup during sustained pushes; the simplest conditional-integration variant is evaluated in Appendix A and broader anti-windup families remain open. Pitch stiffness itself is fixed at k_p=50, since the anchor and boost resolve the precision and push trade-off without gain scheduling, and a scheduled variant softening k_p from 50 to 35 during pushes was tested and rejected because it did not improve push survival when S0 and S1 are compared in Table 3.

4.3. Proximity-Gated Anchor Mechanism ($g_{\mathbf{anchor}} \cdot \boldsymbol{\tau}_{\mathbf{anchor}}$)

The anchor mechanism is the core contribution. It adds three terms to the wheel-torque channel which are active only when the robot is near home and genuinely at rest,

$$\begin{aligned}\boldsymbol{\tau}_{\text{anchor}} = K_i \int g_{\text{prox}} \cdot g_{\text{env}} \cdot (-\Delta x)\, dt \\ + g_{\text{boost}} \cdot K_{\text{boost}} \cdot (-v_{\text{sag}}) \\ + \left(k_{d,h} g_h + k_{d,a} g_{\text{prox}}\right) g_{\dot\theta} \cdot (-\dot\theta),\end{aligned} \tag{8}$$

where every gate is a continuous smoothstep designed to equal one when the robot is at home and quiet and to fall smoothly to zero otherwise,

$$g_{\text{prox}}(\Delta x) = 1 - \text{ss}\big(|\Delta x|;\ 0.05\,\text{m},\ 0.15\,\text{m}\big), \tag{9}$$

$$g_{\text{env}}(v_{\text{sag}}) = 1 - \text{ss}\big(\text{EMA}(|v_{\text{sag}}|);\ 0.25\,\text{m/s},\ 0.50\,\text{m/s}\big), \tag{10}$$

$$\begin{aligned}g_\theta(\theta, \dot\theta) = \big[1 - \text{ss}(|\theta|;\ 2°,\ 12°)\big] \\ \times \big[1 - \text{ss}(|\dot\theta|;\ 2\,°/\text{s},\ 15\,°/\text{s})\big],\end{aligned} \tag{11}$$

$$g_h(\Delta h) = 1 - \text{ss}\big(|\Delta h|;\ 0.05\,\text{mm},\ 0.30\,\text{mm}\big), \tag{12}$$

$$g_{\dot\theta}(\dot\theta) = \text{ss}\big(|\dot\theta|;\ 2\,°/\text{s},\ 15\,°/\text{s}\big), \tag{13}$$

$$g_{\text{boost}} = g_{\text{prox}} \cdot g_{\text{env}} \cdot g_\theta, \tag{14}$$

where $\text{ss}(x; a, b)$ denotes the smoothstep of Eq. (2) with lower and upper knees a and b, so each gate is dimensionless, monotone and C^1, equal to one inside its home-and-quiet band and falling smoothly to zero outside it. Three physically distinct conditions must therefore hold at once for the anchor to integrate, the robot being spatially near home through g_{prox}, temporally quiet through g_{env} and attitudinally settled through g_θ. The pitch stability gate closes when either pitch or pitch rate leaves its band, which prevents the damping boost from fighting a gentle sustained push that never triggers the velocity envelope, and saturates at one when the robot is genuinely at rest. The third anchor term is a gated pitch-rate term whose gate $g_{\dot\theta}$ is the complement of the rate factor in g_θ, so it contributes nothing in quiet stance and engages only once the pitch rate leaves that band while the robot is still near home and still tracking its commanded height, applied common-mode to both wheels.

Reliable quiet-stance detection rests on an asymmetric exponentially weighted moving average of $|v_{\text{sag}}|$,

$$\text{EMA}_t = \begin{cases} \alpha_a |v_t| + (1-\alpha_a)\text{EMA}_{t-1}, & \text{if } |v_t| > \text{EMA}_{t-1}, \\ \alpha_r |v_t| + (1-\alpha_r)\text{EMA}_{t-1}, & \text{otherwise,} \end{cases} \tag{15}$$

whose primary specification is the pair of time constants $\tau_{\text{attack}} \approx 23$ ms and $\tau_{\text{release}} \approx 1.5$ s. The discrete coefficients follow from $\alpha_a = 1 - \exp(-\Delta t/\tau_{\text{attack}}) \approx 0.35$ and $\alpha_r = 1 - \exp(-\Delta t/\tau_{\text{release}}) \approx 0.007$ at 100 Hz. The asymmetry is essential because a symmetric EMA is bistable, post-push oscillation holding it in a middle band where the gate stands half open and the boost cannot collapse the cycle, an outcome confirmed by the symmetric-EMA ablation of Section 5. The fast-attack and slow-release envelope instead rises to one only when oscillation truly decays and falls to zero within roughly 25 ms of a disturbance. The mechanism is a peak-hold envelope detector with asymmetric time constants, and ACC's contribution is to couple that detector to a spatial proximity gate for integral conditioning, the complete activation sequence through a push-recovery cycle being shown in Fig. 3. One implementation detail follows from it, namely that starting the envelope at zero biases g_{env} toward one for about 7.5 s before velocity noise accumulates and the bias decays, so measurements allow 3 – 5 s of settling and a hardware implementation should initialize the envelope at unity instead.

4.4. Posture and Yaw Stability ($\boldsymbol{\tau}_{\mathbf{posture}}$)

Posture regulation uses Jacobian-based PD gains scheduled across twenty-three heights,

$$\begin{aligned}\boldsymbol{\tau}_{\text{posture}} = \boldsymbol{\tau}_{\text{PD}}^{\text{jac}}(q, \dot q, h^{\text{cmd}}) + \boldsymbol{\tau}_{\text{div}}(q_{\text{hy}}^L, q_{\text{hy}}^R) \\ + g_{\text{settled}}\, \boldsymbol{\tau}_{\text{homing}}(q - q_{\text{ref}}),\end{aligned} \tag{16}$$

where $\boldsymbol{\tau}_{\text{div}}$ damps hip-yaw divergence and $\boldsymbol{\tau}_{\text{homing}}$ restores nominal posture when upright and settled. Yaw return uses wheel-differential torque.

4.5. Contact-Loss Recovery ($g_{\mathbf{flight}} \cdot \boldsymbol{\tau}_{\mathbf{flight}}$)

When both wheels lose ground contact, ACC detects the event through per-wheel normal forces and substitutes a flight attitude controller that uses the wheels as reaction flywheels,

$$g_{\text{flight}} = \begin{cases} 1, & \text{if } F_z^L, F_z^R < 0.5mg \text{ for } \geq 2 \text{ steps,} \\ 0, & \text{if } \min(F_z^L, F_z^R) \geq 0.5mg \text{ for } \geq 5 \text{ steps,} \end{cases} \tag{17}$$

$$\boldsymbol{\tau}_{\text{flight}} = \text{clip}\big(2.0\theta + 0.5\dot\theta - 0.02\omega_{\text{wheel}},\ \pm 2\,\text{Nm}\big). \tag{18}$$

Forward wheel spin pitches the torso nose-up during descent, and the five-step release hysteresis prevents balance torque from re-engaging during bounce. The free-fall drop

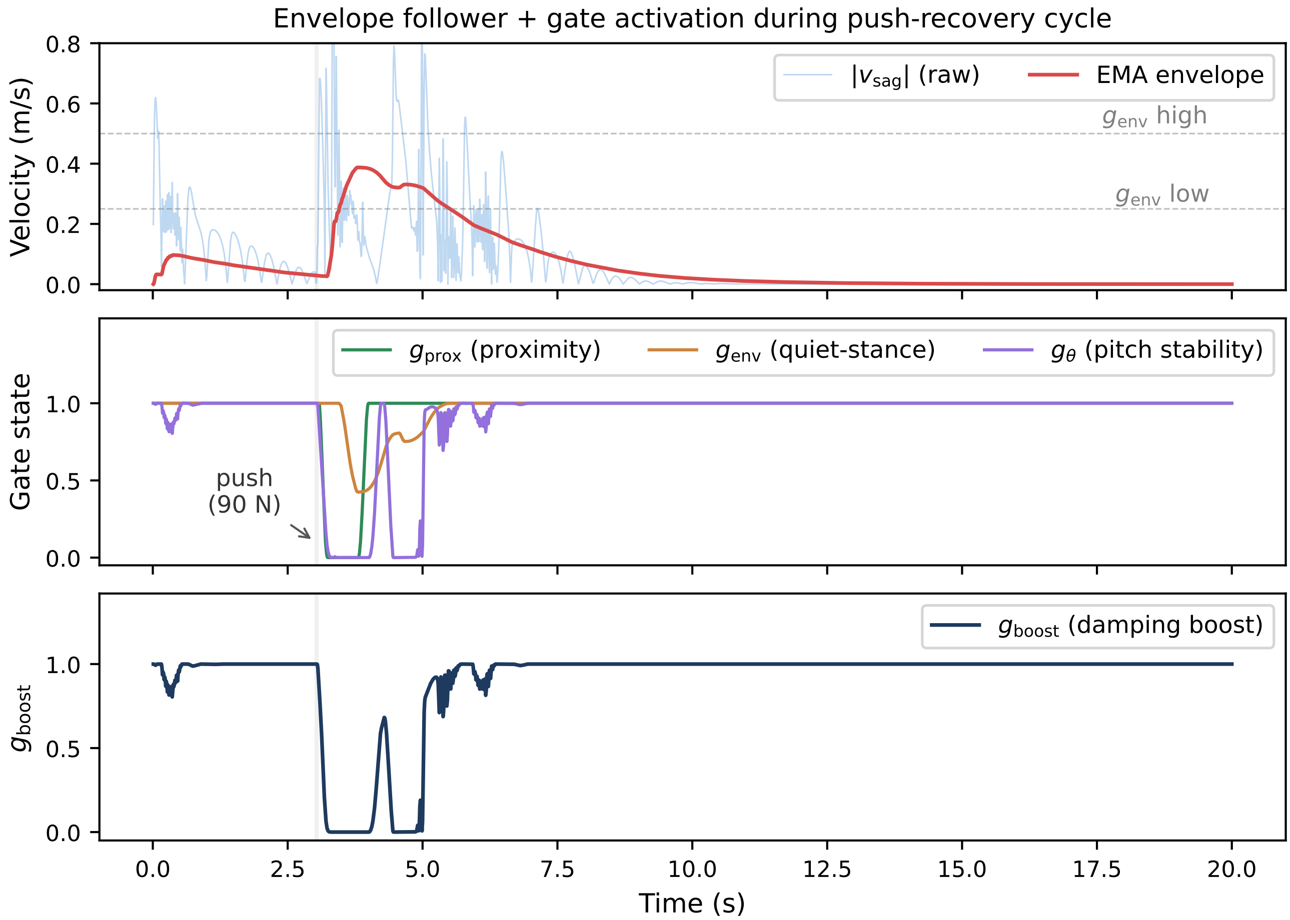

Figure 3: Gate activation during a 90 N lateral (world $+x$) push-recovery cycle (100 Hz control). **Top:** $|v_{\text{sag}}|$ and asymmetric EMA envelope; dashed lines mark the 0.25 – 0.50 m/s quiet-stance band. **Middle:** Individual gate states. **Bottom:** Composite damping boost g_{boost}. Grey shading: 7-step push window. The three panels are the empirical basis for finding (iii) of Section 5.1.5: g_{prox} and g_{env} both sit pinned at their quiet-stance values before the impulse and after the ringdown, which is why ablating either reproduces ACC bit-for-bit at idle, and why only a disturbance experiment can measure their contribution.

is a stress test while the ledge drive-off is the intended use case. Once the wheels are unloaded the wheel–inverted-pendulum loop that this override replaces is itself a reaction-flywheel controller, so the override's measured contribution is to bound landing-attitude variance near the upper end of the drop envelope rather than to enable recovery at all, a distinction Section 5.3 separates over ten trials.

4.6. Per-Leg Ground Adaptation (g_{terrain})

When the two wheels rest at different ground heights, as they do on a curb straddle or an oblique ledge, a single torso height command forces the torso to roll by the step angle. ACC instead maintains an independent ground estimate per leg and issues two height commands. The estimate is taken from the wheel contact-point height, which is exact at any body tilt for a rigid wheel, and is updated only while that wheel carries real load,

$$\hat{h}^i_{\text{gnd}} \leftarrow \hat{h}^i_{\text{gnd}} + \alpha_g \left(z^i_{\text{contact}} - \hat{h}^i_{\text{gnd}}\right), \; F^i_z \geq 0.2mg, \; i \in \{L, R\}, \tag{19}$$

with $\alpha_g = 0.35$ per control step to reject contact chatter. Testing for $0.2mg$ of load rather than for mere geometric touching is deliberate, because a lightly touching wheel in the act of unloading rides a few centimetres high and injected a 2.9 cm phantom ground step on the 15 cm curb. An unloaded wheel therefore freezes its estimate rather than updating Eq. (19), since an airborne wheel supplies no information about where its ground is, the one exception being the only physically unambiguous signal that the ground has vanished, namely the wheel centre descending below its own believed ground,

$$\delta^i > 3\,\text{mm} \;\Rightarrow\; \dot{\hat{h}}^i_{\text{gnd}} = -\left(v_0 + k_s\,\delta^i\right), \tag{20}$$

where $\delta^i = \hat{h}^i_{\text{gnd}} - z^i_{\text{contact}}$ is the depth by which the wheel has fallen below its believed ground, with $v_0 = 0.5$ m/s and $k_s = 50\,\text{s}^{-1}$. The proportional term is what makes the seek usable, since a fixed slew converged the estimate only after 0.30 s, by which time a 20 cm dismount had already tipped the robot past recovery, whereas a 15 cm drop slews at 8 m/s and converges within two control steps, so the leg extends during the fall rather than after it. When both wheels are airborne both estimates track the mean wheel height, which reproduces the symmetric-posture behaviour exactly and leaves ballistic flight to Section 4.5.

The step $\Delta h = \hat{h}^{L}_{\text{gnd}} - \hat{h}^{R}_{\text{gnd}}$ is then absorbed by the legs rather than by the torso, and the split is kinematic rather than a linear height offset. Writing $L(\cdot)$ for the forward-kinematic leg length and L^{-1} for its inverse over the calibrated posture range, the leg on the higher ground is shortened by the full step,

$$L_{\text{high}} = L\big(h^{\text{cmd}}\big) - |\Delta h|, \tag{21}$$

$$h^{\text{high}} = L^{-1}\big(L_{\text{high}}\big), \qquad h^{\text{low}} = h^{\text{cmd}}, \tag{22}$$

both clamped to the calibrated envelope extended by 10 cm at each end. To first order this reduces to the familiar symmetric split $h^{\text{cmd}} \mp \Delta h/2$, but the map from height to leg length is markedly nonlinear near full squat and the linear proxy erred by 12 cm at the lower extrapolation limit, folding the knee back under the thigh. Should L_{high} fall below the shortest physically reachable leg, ACC raises the whole robot until the short leg fits, trading commanded height for the ability to span the step, though a 20 cm curb at nominal stance exceeds even that remedy since it would demand 170° of knee flexion against a 2.7 rad limit. The uncompensated remainder then appears as a predicted torso roll of $\arctan(-\epsilon/w_{\text{track}})$ for an unspanned step ϵ and wheel separation w_{track}=0.278 m, and reporting that expected roll explicitly is what keeps a legitimate straddle from being misread as an incipient fall by the tilt-based safety monitors.

Because the height-to-length table is itself extrapolated at the extremes, a purely feedforward split left a systematic 6° torso roll on the 10 cm curb. A closed-loop leveling integrator therefore trims the split by the measured roll at 0.15 m/(rad s) with ±6 cm of authority, integrating only when both wheels are loaded, a genuine ground step of at least 2 cm is present and the ground estimate is settling more slowly than 0.15 m/s, and decaying to zero with a 1 s time constant away from a step. That gating is not cosmetic, since without the step condition the dynamic roll of a turn is integrated as terrain and wound an 8.6° roll bias through a 180° turn, while without the rate condition the geometric transient of a curb climb is integrated as a leveling error and compresses the flat leg.

The expected roll must also reach the gates and not only the posture solver, because heading hold is a differential wheel torque scaled by a stability envelope whose roll factor decays over 1 – 5° while the unspanned remainder of a 20 cm step is a quasi-static 2.5 – 4.0°, so the envelope read a legitimate straddle as an incipient fall, suppressed heading hold for the entire crossing and left the mount transient undamped until the robot yawed off the slab. ACC therefore widens the roll band in proportion to the commanded split, over the same 3 – 10 cm interval the wheel loop already uses, and restores both the differential term and the mean-ground height reference once the straddle has settled, settling being decided by the rate of the commanded split against a 0.2 s exponential average, which separates cleanly by a factor of ten since $|\Delta\dot{h}|$ runs at 0.05 – 0.13 m/s while a wheel climbs and below 0.016 m/s once astride. Restricting the relaxation to the settled phase is a requirement rather than conservatism, because applying it while a wheel is still crossing an edge keeps drift authority alive over the drop and topples the robot on a 45° oblique exit. On flat ground the step vanishes identically and every term above with it.

Table 1
Core numerical parameters for the ACC balance controller. All values in SI units unless noted otherwise.

Balance Core			**Anchor Mechanism**		
k_p	50	Nm/rad	K_i	4.0	Nm/(m s)
k_d	10.0	Nm s/rad	Integral cap	2.0	Nm
k_{pos}	40	Nm/m	Integral leak	5×10^{-3}	per step
Position clip	±4.0	Nm	k_{vb} (boost)	5.0	—
k_{vel}	15	Nm s/m	$k_{d,h}$ / $k_{d,a}$	3.0 / 7.0	Nm s/rad
Damp. scale s_v	1.5	—	g_h band	0.05–0.30	mm
k_{v2}	5.0	Nm s/m	g_{prox} band	0.05–0.15	m
k_w (per-wheel)	0.5	Nm s/rad	g_{env} band	0.25–0.50	m/s
k_{roll}	40	Nm/rad	τ_{attack}	0.023	s
$k_{\text{roll},d}$	8.0	Nm s/rad	τ_{release}	1.5	s
k_{yaw}	8.0	Nm/rad	g_{stab} (θ)	2–12	°
$k_{\text{yaw},d}$	2.0	Nm s/rad	g_{stab} $(\dot{\theta})$	2–15	°/s
$k_{\text{pos,return}}$	2.0	Nm/m	**Flight / Terrain**		
k_{heading}	2.0	Nm/rad	k_{flight} (P / D)	2.0 / 0.5	—
$k_{\text{heading},d}$	0.6	Nm s/rad	Wheel damp. (fl.)	0.02	Nm s/rad
			Engage / release	2 / 5	steps
			F_z thresh.	0.5 / 0.2	$\times mg$

4.7. Complete Numerical Parameter Set

Table 1 lists the core parameters, and the full set of more than fifty gains is available in the supplementary material.

5. Experimental Validation

ACC is evaluated across four scenario classes, each mapped to the component it tests and paired with targeted ablations. ACC and all six classical baselines run at 500 Hz physics with 100 Hz control under a 400 Nm/s torque rate limit per joint, so the headline comparison is rate-matched by construction, while the PPO reference is reported at the 50 Hz rate its policy was trained at. The baselines were additionally measured at the 50 Hz rate for which their gains were tuned, and since the fall rate is total in all twelve cells of Section 5.4.1 the comparison does not turn on which rate is chosen. The converse direction is a negative result, because recomputing ACC's discrete coefficients for the longer step gives a controller that is not preserved, the robot collapsing to a mean CoM height of 0.124 m against a nominal 0.404 m, so that the 0.050 mm idle RMS and 10 N push threshold the run produces are artifacts of a machine lying motionless on the ground. No 50 Hz ACC result is therefore reported and rate-independence in the downward direction is not claimed, an issue taken up as Limitation 2. Seeds are deterministic throughout, idle standing using seed=20260727+trial_id, the push ablation seed=20260728×100+rep and the LQR baselines the seeds $\{0, 42, 123\}$, and every behaviour below that has a visible time course is also supplied as video in Section 5.8.

5.1. Anchored Standing

5.1.1. Two precision metrics, deliberately kept separate

"Standing precision" names two different quantities in the wheeled-biped literature, and they are not interchangeable. Let $\mathbf{p}(t) \in \mathbb{R}^2$ be the horizontal base position over an evaluation window of length T, let $\mathbf{p}_0$ be the commanded home position, and let $\bar{\mathbf{p}} = \frac{1}{T}\int_0^T \mathbf{p}(t)\,dt$ be the trial mean. We report

$$A = \left(\frac{1}{T}\int_0^T \|\mathbf{p}(t) - \mathbf{p}_0\|^2\,dt\right)^{1/2}, \tag{23}$$

$$R = \left(\frac{1}{T}\int_0^T \|\mathbf{p}(t) - \bar{\mathbf{p}}\|^2\,dt\right)^{1/2}. \tag{24}$$

The first is the accuracy, an RMS displacement from the position the controller was told to hold, which retains any DC parking offset and answers how close to the commanded pose the robot parks. The second is the repeatability, an RMS about the trial's own mean with the DC offset removed, which answers how still the robot is once it has parked. A controller with a persistent 27 mm offset and a perfectly steady hold scores A=27 mm at R≈0 mm while a controller centred on the target but oscillating scores the reverse, so reporting either alone is uninformative and, since the two differ by more than an order of magnitude here, comparing one against the other produces a meaningless ratio.

Both are reported for every configuration below and resolved separately along the sagittal axis, the lateral axis and the full plane, because the two horizontal axes of this platform differ in kind rather than in degree and must not be pooled. The convention is fixed explicitly here, since the two are easy to transpose and the physical conclusions invert if they are. Both wheel axles point along world x and the wheels are separated along it by w_{track}=0.278 m, so the sagittal axis is world y, which is where wheel torque produces ground force and the only axis carrying a genuine inverted-pendulum mode, while the lateral axis is world x, where no actuator produces ground force and the axis is held passively by the wheel contacts and by the roll restoring moment of the track width. Two dynamic checks confirm the assignment rather than merely asserting it, since a 90 N impulse along y is absorbed at a peak wheel speed of 52 rad/s and leaves only 4.8 mm of residual offset as the wheels drive back under the centre of mass, whereas the same impulse along x leaves a permanent 23.2 mm offset, and over a 20 s quiet stance the robot rolls 24.6 mm along y with the wheel midpoint tracking the centre of mass to 0.2 mm while x holds to 0.02 mm.[1] The convention matters for reading Table 2, because the anchor integral and the gated damping boost both live in the wheel-torque channel and therefore act on the sagittal axis alone, so the lateral column measures how little the plant moves when nothing drives it rather than any property of the controller.

[1]All four checks are reproduced by `scripts/verify_body_axes.py`, which asserts on disagreement.

5.1.2. Unified measurement protocol

Every row of Table 2 is measured under one protocol. Each trial runs 25 s at 500 Hz physics and 100 Hz control from the nominal standing pose under randomized initial conditions, with σ=0.005 rad per joint clipped to the joint limits and σ=1 mm on root height, over ten trials. The first 5 s are discarded as settling, which is long enough to cover the transient of an envelope initialized at zero since that transient comes within one percent of the quiet-stance value in under 5 s, and A and R are computed over the remaining 20 s relative to each trial's own pose at t=0, so $\mathbf{p}_0$ is the commanded home rather than the world origin. A trial is scored as a fall if $|\theta|$ exceeds 46° or the root height drops below 0.15 m, and confidence intervals are Student-t. The protocol carries one scope limit, in that starting every trial at the commanded home never exercises the anchor's return-to-home function, so it measures the controller's ability to stay put rather than to come back, the latter being measured by the push experiments of Section 5.2.

5.1.3. Result

ACC holds sub-millimetre sagittal repeatability at R_{sag} = 0.294±0.015 mm, with a 95% confidence interval of [0.283, 0.305] mm across ten randomized trials and no falls. This is the figure of merit for the platform, because the sagittal direction is the actuated inverted-pendulum axis. It is the axis on which the robot can fall, the axis the wheel-torque channel drives, and the axis every configuration in the ladder is trying to hold.

Signal-to-noise ratios against a pinned-base simulator noise floor are not reported, since pinning zeroes velocities without welding the base and the floor it yields is meaningful only on the laterally constrained axis at 0.0019 mm RMS rather than on the gravity-loaded ones. Precision is therefore assessed from the relative spread between configurations measured under an identical protocol. Removing the gated damping boost alone in configuration S2 moves R_{sag} from 0.294 to 4.772 mm, a factor of sixteen with non-overlapping standard deviations, which establishes that the metric resolves the effect under study.

Measured against the P-only baseline L0, which sustains the 69.7 mm peak-to-peak sagittal limit cycle described in Section 1, the like-for-like improvements are a factor of 59.4 in sagittal repeatability from 17.461 to 0.294 mm, a factor of 22.6 in planar repeatability from 17.845 to 0.788 mm, and a factor of 5.0 in lateral repeatability from 3.675 to 0.730 mm. The ordering of these three factors follows the actuation structure and is the signature of a wheel-torque controller, because every term ACC adds beyond L0, namely position homing, the anchor integral and the gated damping boost, produces ground force along the rolling direction and none can produce lateral force at all, so the residual fivefold lateral improvement is an indirect consequence of the robot pitching and yawing less rather than of any lateral control action. The sagittal figure is therefore the axis-resolved one for this class of platform, while the lateral figure and any planar norm are diluted by an axis the controller does not actuate.

Table 2
Idle standing precision across the full ablation ladder. All rows measured under the single protocol of Section 5.1; values are mean±SD over N=10 trials. See the note below the table for column definitions.

	Configuration	R_{sag} (mm) sagittal rep.	R_{xy} (mm) planar rep.	R_{lat} (mm) lateral rep.	$\bar{s}$ (mm) sag. offset	A_{lat} (mm) lateral acc.	Survival
Additive ladder (bottom-up)							
L0	P-only (PD, no position, no integral)	17.461 ± 0.258	17.845 ± 0.279	3.675 ± 0.246	−44.8	9.933 ± 0.712	10/10
L1	+ position homing (±0.10 m)	20.397 ± 0.017	20.414 ± 0.019	0.834 ± 0.066	−30.4	2.629 ± 0.884	10/10
L2	+ ungated integral K_i	25.026 ± 0.039	25.056 ± 0.043	1.212 ± 0.110	−27.5	3.661 ± 1.264	10/10
L2.5	+ proximity gate only (no envelope, no boost)	25.198 ± 0.017	25.227 ± 0.020	1.205 ± 0.099	−27.6	4.023 ± 1.194	10/10
Proposed controller							
L3/S1	**ACC** (full anchor, fixed k_p=50)	$\mathbf{0.294 \pm 0.015}$	$\mathbf{0.788 \pm 0.069}$	$\mathbf{0.730 \pm 0.073}$	**−27.0**	$\mathbf{1.213 \pm 0.554}$	**10/10**
Subtractive ablation (top-down from ACC)							
S0	ACC + scheduled k_p (rejected variant)	0.294 ± 0.015	0.788 ± 0.069	0.730 ± 0.073	−27.0	1.213 ± 0.554	10/10
S2	− gated damping boost	4.772 ± 0.069	4.824 ± 0.067	0.703 ± 0.069	−26.8	1.260 ± 0.616	10/10
S3	− asymmetric envelope (symmetric EMA)	0.294 ± 0.015	0.788 ± 0.069	0.730 ± 0.073	−27.0	1.213 ± 0.554	10/10
S4	− proximity gate (integral always on)	0.294 ± 0.015	0.788 ± 0.069	0.730 ± 0.073	−27.0	1.213 ± 0.554	10/10
S5	− pitch safety gate g_θ	—	—	—	—	—	**0/10**
Single-mechanism isolation (ACC with one term disabled)							
M1	K_i=0 (anchor integral removed, gates intact)	0.089 ± 0.035	0.733 ± 0.072	0.727 ± 0.071	−31.2	1.192 ± 0.535	10/10
M2	damping terms reverted to their L2 values	25.198 ± 0.017	25.227 ± 0.020	1.205 ± 0.099	−27.6	4.023 ± 1.194	10/10
M3	K_i=0 *and* boost = 0	3.973 ± 0.130	4.037 ± 0.128	0.710 ± 0.069	−31.1	1.241 ± 0.591	10/10

N=10 trials, 100 Hz control, 20 s measurement window after a 5 s settle, clean sensors and zero actuator delay. R is repeatability, the RMS about the trial mean of Eq. (24), A is accuracy, the RMS from commanded home of Eq. (23), and $\bar{s}$ is the mean signed sagittal parking offset, sagittal being world y and lateral world x. Because the anchor integral and the damping boost act on the sagittal axis only, R_{sag} and $\bar{s}$ are the discriminating columns while R_{lat} measures a passively held axis and is near-constant by construction. A_{sag} and A_{xy} are omitted because they add nothing, the identities $A_{\text{sag}} = \sqrt{\bar{s}^2 + R_{\text{sag}}^2}$ and $A_{xy} = \sqrt{A_{\text{sag}}^2 + A_{\text{lat}}^2}$ holding to 0.003 mm and 0.02 mm over all twelve surviving rows. Produced by `scripts/collect_idle_ladder.py`; raw output `outputs/paper_verification/idle_ladder.json`.

Accuracy and repeatability remain distinct quantities on that axis and both are reported. ACC parks at $\bar{s} = -27.0$ mm from the commanded home and holds that standoff to within 0.009 mm of standard deviation across trials, so the settle terminates at a fixed offset rather than at the commanded point, and planar accuracy is accordingly $A_{xy} = 26.994$ mm, a 1.8-fold improvement on L0 against the 59-fold gain in repeatability. The offset lies on the axis the anchor integral does control, since the integral moves it from 31.2 mm in configuration M1 to 27.0 mm, and its magnitude is fully accounted for. Appendix C instruments the sagittal torque balance and predicts it in closed form from two settings, a leaky anchor integral whose 2 s forgetting horizon fixes its DC gain at the finite $k_{I,\text{dc}}$=7.96 Nm/m, so that a constant load leaves a finite error by construction, together with a 1.44° error in the feedforward pitch reference. The prediction is 24.88 mm against 24.99 mm measured and holds to 0.92 % across a fortyfold gain sweep and a five-point height sweep. The leak also buys the steadiness, since a tenfold smaller λ cuts the offset to −16.31 mm but degrades window jitter twelvefold, so what Table 2 publishes is a quantified and deliberately chosen point on that trade-off with both alternatives priced against the full evaluation suite.

5.1.4. The measurement window is a protocol choice, not a stability horizon

Every row of Table 2 is measured over 20 s, and Section 5.4.2 declines to promote a competing design specifically because a 20 s horizon flatters it. The ACC row was therefore re-run with identical seeds, initial conditions, profile and metrics and only the window lengthened, over 300 s and 1800 s. Neither extension produces a fall, at 10/10 and 4/4 respectively, and sagittal repeatability improves from the published 0.294±0.015 mm over 20 s to 0.089±0.005 mm over 300 s and 0.037±0.002 mm over half an hour, while the least-squares sagittal drift rate falls from 0.0149 mm/min to 0.0004 mm/min. The 20 s column reproduces the published figure exactly, so these are the same measurement at three window lengths.[2]

Three readings follow. The headline 20 s figure is the conservative one, since that window still contains the tail of the settle, so lengthening it improves repeatability by 3.3 and 7.9 times rather than degrading it. The settle also terminates in a fixed point rather than in a slow mode, since after roughly 35 s the sagittal coordinate is constant to 2×10^{-6} mm per 30 s block for the remaining twenty-nine minutes, the parking offset holding at −26.763 mm and CoM height at 0.4068 m, and the only motion left anywhere in the state is a monotone lateral creep of 0.14 mm/hour on the passively held axis, four orders of magnitude below the standoff it would have to traverse to matter. The separation from the full-state LQR of Section 5.4.2 is finally qualitative rather than a matter of degree, because under the same extension that design accumulates 0.43 – 0.63 m of planar drift and falls between 22 s and 57 s, whereas ACC's position error is referenced to a

[2] 100 Hz control, 500 Hz physics, 5 s settle before the window, clean sensors, zero delay, seeds 20260727+i, profile `V3_ANCHOR`. Produced by `scripts/idle_longhorizon.py`; raw output `outputs/paper_verification/idle_longhorizon_\{300,1800\}s.json`.

latched home by two explicit terms, the clipped proportional τ_{pos} of Eq. (3) and the anchor integral on $-\Delta x$ of Eq. (8), and terminates at the 27.0 mm standoff that Appendix C derives in closed form. What Table 2 reports is therefore a window chosen to match the baselines, not the longest interval over which the claim holds.

5.1.5. *Which mechanism produces the precision*

The bottom two blocks of Table 2 isolate the contributions, and four findings are worth stating precisely because three of them contradict the attribution that the gate structure invites.

The first is that idle steadiness comes from the damping ladder while the integral buys accuracy instead. M1 disables the anchor integral entirely with K_i=0 while leaving every gate in place, and its repeatability is not merely as good as ACC's but slightly better at 0.089 mm against 0.294 mm sagittally, while its accuracy is worse by 4.2 mm, parking at $\bar{s} = -31.2$ mm against ACC's -27.0 mm. That is the integral's entire measurable idle signature and exactly the trade one expects, since the term walks the robot closer to the commanded home at the cost of a little steadiness, being itself a slowly-varying command, so a protocol reporting repeatability alone would conclude wrongly that the integral is inert at idle while one reporting accuracy alone would miss that it costs a factor of three in steadiness. Conversely M2 keeps the integral and every gate but reverts the two velocity-damping coefficients to their L2 values and reproduces the L2 limit cycle exactly at R_{sag}=25.198 mm. Steadiness is therefore attributable to the damping terms in two stages, since raising the velocity-damping scale takes R_{sag} from 25.2 mm to 3.97 mm with the boost still off in M3 and enabling the gated boost takes it to 0.294 mm, a combined factor of 86 decomposed into roughly 6.3 and 13.5, and S2 confirms the second stage from the top down by degrading R_{sag} to 4.772 mm when only the boost is removed.

The second finding is that the damping boost acts sagittally, as it must, since S2 degrades sagittal repeatability by a factor of 16 while leaving lateral repeatability statistically unchanged at 0.703 mm against ACC's 0.730 mm. Only that outcome is consistent with the architecture, because the boost multiplies a velocity-damping coefficient inside the wheel-torque channel of Eq. (3) and wheel torque generates ground force along the rolling direction alone, so the residual oscillation it suppresses is a sagittal pitching mode. The distinction matters because the lateral column reads as the plausible x-axis RMS to report by default, yet it is the one column in Table 2 nearly blind to every mechanism in the ladder, spanning 0.70 – 3.68 mm across all twelve surviving configurations and only 0.70 – 0.73 mm across the seven rows that retain the full damping ladder.

The third finding is that the proximity gate and the asymmetric envelope are inert during quiet stance, S4 and S3 being bit-identical to ACC on every idle metric, which is the expected behaviour rather than a null result. The base never leaves the 5 cm inner band, so g_{prox} stays at unity, and the velocity envelope never rises into the attack branch, so the asymmetric and symmetric followers produce the same signal. Both mechanisms are defined by what they do when the robot is displaced, namely windup prevention during a push and ringdown after one, so the idle protocol is blind to both by construction and they are correctly evaluated only in Table 3.

The fourth finding is that the pitch safety gate is required even to stand, since S5 falls in 10/10 trials during pure quiet stance with no push applied, removing g_θ leaving the damping boost free to act on the pitch state at large excursions so that the resulting positive feedback diverges from the ± 0.005 rad initial perturbation alone. The gate is therefore a stability precondition of the quiet-stance configuration itself and not only a disturbance-time safeguard.

Taken together, the anchor's gates make an aggressive damping and integral configuration admissible without windup or divergence while the damping terms they admit deliver the steadiness, so the gates are enabling conditions rather than the proximate cause of the precision. The integral's primary role is drift rejection under disturbance, which by construction cannot appear in a protocol that starts at the target, and its measurable idle signature is confined to the 4.2 mm of parking offset it removes at a threefold cost in sagittal repeatability.

5.2. Omnidirectional Push Recovery

Push recovery is evaluated by a polar sweep in which impulses of 10 – 160 N lasting 7 control steps are applied at the torso along 8 directions. Bearings are applied in the world frame as $\mathbf{f} = F(\cos\theta, \sin\theta, 0)$, so that $\theta = \pm 90°$ are the sagittal bearings and θ=0° and 180° the lateral ones under the axis convention of Section 5.1. We write F_{min} for the maximum magnitude survived in every direction and F_{med} for the median, both located by binary search over [10 N, 160 N] with 8 iterations per direction and a 5 N tolerance. A trial counts as survived when torso pitch stays below 46° for 17 s after the impulse.

The envelope is anisotropic by construction. The two lateral bearings average 98.0 N against 130.0 N for the two sagittal ones, and the weakest bearing of all is the oblique 45° one at 82.1 N, some 24 % below F_{med}. The asymmetry follows from the morphology rather than from tuning, because the wheels can drive the contact patch back under the CoM along the sagittal axis but produce no lateral ground force at all, so lateral recovery falls to the leg and torso alone. The oblique bearings are weakest because neither mechanism has full authority there, and raising the lateral bearing therefore requires a centralized lateral task of the kind discussed in Section 5.5 rather than a gain change.

The envelope is also chiral, and the chirality lives in the controller. Superposed on the structural anisotropy is a left–right asymmetry in which 135° survives 40.7 N more than −135° (Fig. 4). The plant cannot produce it, since mirroring the model about the sagittal plane reproduces its own mass, inertia and geometry to 0.05 mm and 1×10^{-7}, leaving the open-loop system mirror-symmetric to numerical precision.

Fitting $F(\theta)=a_0+a_1\cos\theta+b_1\sin\theta+a_2\cos 2\theta+b_2\sin 2\theta$ localizes the violation, because a mirror symmetry forbids b_1 and b_2 while a 180° rotation forbids a_1 and b_1, and the measured envelope has b_1=0.3 N but b_2=− 17.3 N, an order of magnitude above the between-episode dispersion, so the closed loop is C_2-equivariant and mirror-broken. Its source is the sign convention of the two differential channels, since feeding the controller a state and its mirror image maps the commanded torque to its own mirror image exactly on hip-pitch, knee and wheel and fails only on hip-roll at 3.275 Nm, exactly twice that channel's own command, and on hip-yaw at 0.526 Nm. Both are assembled antisymmetrically as $\tau_L=+m$ and $\tau_R=-m$, the right structure for a differential command but even rather than odd under the mirror map, so the pair commands same-signed torques where the mirror requires opposite-signed ones. Two control experiments close the loop causally, mirroring those channels flipping b_2 from −16.2 N to 15.6 N at unchanged magnitude while a sham mirror that permutes the same quantities without touching the sign convention leaves it at −15.7 N, and inverting the hip-roll sign alone collapses b_2 to 1.2 N. We report the effect without adopting a correction, because removing the chirality redistributes the envelope without enlarging it, F_{min} moving from 82.5 N to 79.7 N in the sign-inverted arm, which makes it a property of this gain set to characterize before porting to a different track width.[3]

Several auxiliary experiments apply a single impulse along world $+x$ rather than repeating the full eight-bearing sweep at eightfold cost, among them the robustness and rate-limit sweeps, the ringdown and gate-activation traces of Figs. 3 and 5, and the compound-disturbance protocol of Section 5.6. That bearing is lateral and below median, the polar sweep placing 0° at 95.5 N, third-weakest of the eight and 12.9 N below F_{med}, so those experiments are conservative probes of ACC's disturbance margin and are labelled lateral throughout rather than forward.

Table 3 reports the factorial ablation.[4] Read additively, position homing adds 5.1 N over the P-only baseline at $p=2\times10^{-23}$, the ungated global integral adds nothing over homing, and the full anchor adds 7.2 N over that integral at $p=1\times10^{-15}$. The step from L2.5 to L3 bundles four mechanisms at once and the push metrics alone cannot separate them, which is why Section 5.1.5 performs the separation on the idle side instead. What the push data do show is that the proximity gate contributes nothing to push margin on its own, consistent with its role being windup prevention rather than capture-basin enlargement, and that the full anchor recovers a margin advantage over the ungated integral while simultaneously being the only configuration in the ladder that settles.

Table 3
Factorial ablation of ACC components: push-recovery and ringdown metrics. Clean sensors, all rows N=30, mean±std. See the note below the table for definitions and significance tests.

	Configuration	F_{min} (N)	F_{med} (N)	Ringdown
Additive (bottom-up)				
L0	P-only (PD, no pos., no I)	70.0±1.3	122.8±3.3	never
L1	+ Pos. homing (±0.10 m)	75.0±1.1	130.5±1.9	never
L2	+ Integral K_i (no gate)	74.9±0.9	131.5±1.4	—
L2.5	+ Prox. gate only (no EMA, no boost)	74.9±0.9	131.4±2.0	—
L3	+ Full anchor (gate, EMA, boost)[7]	**82.1±2.7**	108.4±6.6	**9.0** s
Subtractive (top-down from ACC)				
S0	ACC + scheduled k_p (rejected)[6]	80.7±2.8	110.2±6.2	9.0 s
S1	**ACC**[6] (canonical, fixed k_p=50)	**82.1±2.7**	108.4±6.6	**9.0** s
S2	−Damping boost	81.2±1.0	115.9±6.2	never[1]
S3	−Asym. EMA (sym. EMA)	81.5±2.6	106.1±6.2	never[2]
S4	−Prox. gate (global I)	80.7±2.8	110.2±6.2	windup[3]
S5	−Pitch gate g_θ	$<10^5$	$<10^5$	never[4]

F_{min} and F_{med} are the minimum and median survivable push over 8 bearings, each located by binary search to 5 N tolerance; idle-precision metrics for the same configurations are in Table 2. [1]Boost necessary for ringdown. [2]Symmetric EMA bistable, post-push oscillation holding it in mid-band. [3]Without the proximity gate the integral accumulates throughout the displacement and saturates its cap, so the robot survives the impulse but does not settle; at idle this ablation is bit-identical to ACC. [4]Removing the pitch gate lets the damping boost act on the pitch state at large excursions and the resulting parametric amplification diverges, additionally failing 10/10 at pure idle with no push applied. [5]Left-censored at the search floor: in 231 of 240 trials the bisection never located a survivable magnitude within [10 N, 160 N], the nine exceptions reaching at most 46.3 N, so both entries are upper bounds rather than measurements. [6]S1 is the canonical configuration; S0, which softens k_p from 50 to 35 during pushes, was tested and rejected. [7]L2→L3 bundles the proximity gate, asymmetric EMA, boost and pitch gate, which S2–S5 isolate subtractively. Welch's unpaired t-test, two-sided, N=30 per group, uncorrected p against the Bonferroni-adjusted α=0.0056 of Section 6: L0→L1 +5.1 N ($p=2\times10^{-23}$, d=4.3) and L2.5→L3 +7.2 N ($p=1\times10^{-15}$, d=3.6) are significant, while L1→L2 and L2→L2.5 are not (p=0.69, 0.86), so the gate alone adds nothing. Every subtractive row lies within 1.4 N of ACC at $|d|\leq0.49$ and none is significant ($p\geq0.061$) except S5 ($p=2\times10^{-44}$). Fig. 4 plots the same 30 repetitions of row S1. Produced by `scripts/replicate_ablation_n30.py`.

Read subtractively, the striking result is how flat the ablation is, since every single-mechanism removal except the pitch gate leaves F_{min} within 1.4 N of ACC and none of them significantly. At N=30 this is a bounded effect rather than an underpowered null, because the four subtractive effect sizes satisfy $|d|\leq0.49$ and the 95 % intervals put any single-mechanism contribution to F_{min} below roughly 1.4 N against the 7.2 N the anchor adds as a whole. Capture basin

[3]Sweep `scripts/sweep_push_chirality.py` with N=5 per bearing over four arms, raw output in `outputs/push_chirality/`. The b_2 of that sweep's shipped arm, −16.2 N, and of row S1 of Table 3, −17.3 N, agree to 1.1 N although the two use different seeds, repetition counts and settle protocols.

[4]`compute_v3_torque_for_state` mutates the controller context it is passed: it carries `prev_com_pos` and the flight-mode latch (`airborne_mode`, `airborne_count`, `ground_count`) there, so every trial in this table is run from a freshly constructed context. Trials are therefore statistically independent, and in particular a trial that ends airborne cannot hand a latched flight mode to its successor in the bisection. Raw outputs, all rows: `outputs/paper_statistics/ablation_n30_results.json`, one harness, one protocol, seeds 20 260 728×100 + r for $r\in[0,30)$.

is set by the anchor collectively and not by any one of its four gates. That conclusion covers S0, since gain scheduling proves unnecessary and fixing k_p=50 costs nothing measurable, and it covers the damping boost, whose value lies elsewhere. S2 is the dominant idle-precision mechanism and acts sagittally, degrading sagittal repeatability sixteen-fold while leaving lateral repeatability statistically unchanged, yet its push margin is indistinguishable from ACC's at 81.2 N against 82.1 N. The boost therefore buys sagittal steadiness and settling rather than capture basin. The one mechanism the push data do isolate is the pitch gate, whose removal leaves the robot unable to survive the 10 N floor of the search in 231 of 240 trials and additionally causes 10/10 falls at pure idle. The asymmetric envelope and the boost are each individually required for ringdown.

5.3. Flight Recovery and Terrain Adaptation

ACC's two-channel assembly supports independent extensions without modifying the core anchor, and both extensions use the same 400 Nm/s rate limit and 100 Hz control rate.

Turning first to contact-loss recovery, ACC returns to a settled stand from a 100 cm free drop in 10 of 10 trials at 24.0±1.2° peak pitch after touchdown, and from a 50 cm ledge drive-off in 10 of 10 at 27.1±2.0°, with the 60 cm drop at 18.2±1.0° and the 20 cm ledge at 19.6±0.3° correspondingly milder.[5] The two flight ablations give a more nuanced picture than the mechanism's design rationale suggests. Removing the flight attitude PD while retaining spin-down damping is a real but modest loss, peak pitch rising to 27.8±0.3° on the 100 cm drop with one fall in ten and to 31.6±2.2° on the 50 cm ledge with none, whereas removing the airborne wheel-torque override entirely, so that the ordinary wheel–inverted-pendulum loop keeps driving the wheels through flight, does not degrade either condition and in fact halves peak pitch to 13.0±0.5° and 10.2±0.4° at full survival, because free wheels driven by the pitch-feedback loop are themselves an effective reaction flywheel. The override's benefit appears only at the upper end of the envelope, where at 150 cm the two configurations survive equally at 2 of 5 but the override holds peak pitch to 34.0±2.1° against 53.6±35.0°, a sixteenfold reduction in spread with one uncontrolled trial exceeding 90°, while at 200 cm both fail. The override therefore bounds landing-attitude variance near the envelope limit rather than enabling the recovery, and at the 50 – 100 cm scales reported here ACC's balance core alone suffices.

The success criterion for terrain adaptation is traversal rather than mere survival, so a trial counts only if the elevated wheel stays on the 36 cm-wide slab for the whole

[5] All at 400 Nm/s rate limit and 100 Hz control, N=10 per condition with randomized initial conditions of ±0.003 rad per joint and ±1.5 mm on the CoM, ± denoting the standard deviation. Survival is the strict settle verdict, namely a flat pitch and roll window, tail height error at most 12 mm and residual speed at most 0.05 m/s, rather than merely staying upright; the drop metric is peak $|\theta|$ from touchdown onward and the ledge metric peak $|\theta|$ through landing, while the curb metric is max $|\phi|$ while straddling, over traversing trials only. Produced by `scripts/collect_flight_terrain_n10.py`; raw output `outputs/flight_terrain_n10/results.json`.

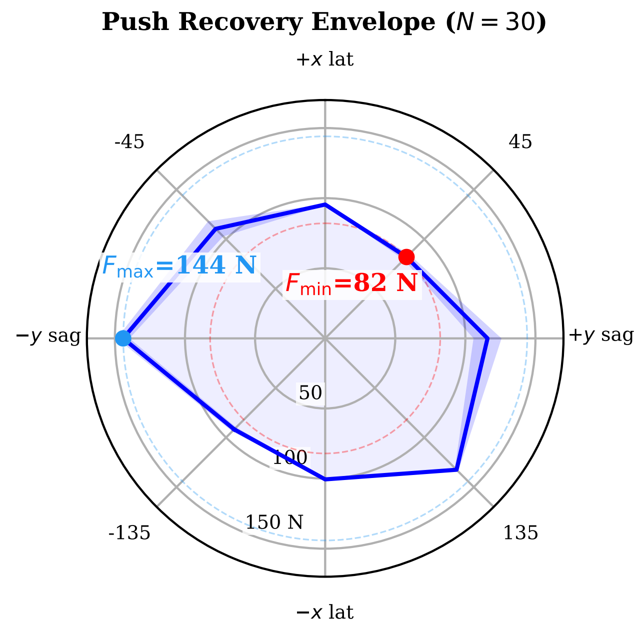

Figure 4: Omnidirectional push envelope of ACC. This is the *same* run as row S1 of Table 3, not a separate measurement: 8 bearings, threshold located by binary search over 10 – 160 N to a 5 N tolerance, so the plotted minimum is by construction the $F_{\min}$ of that row. F_{med} is not read off this figure: it is the per-repetition median over the 8 bearings, averaged over repetitions (108.4 N), which differs from the median of the per-bearing means plotted here (105.6 N) because median and mean do not commute. Solid curve: per-bearing mean survivable push magnitude; shaded band: ±1 standard deviation over the 30 repetitions. The radial axis is linear in force and originates at 0 N, so radial distance is proportional to margin. The envelope is strongly anisotropic, spanning 82.1 – 144.1 N, a 1.8× ratio between the weakest and strongest bearing. The ordering is −90° (144.1 N) > 135° (132.5 N) > 90° (115.8 N) > −45° (110.5 N) > 180° (100.6 N) > 0° (95.5 N) > −135° (91.8 N) > 45° (82.1 N). Two regularities are visible. First, the actuated axis is the strong one: bearings are applied in the world frame as $\mathbf{f} = F(\cos\theta, \sin\theta, 0)$, so ±90° are purely sagittal and 0°/180° purely lateral (Section 5.1). The two sagittal bearings average 130.0 N against 98.0 N for the two lateral ones, a 1.33× advantage. This is the expected asymmetry for a wheeled biped: sagittally the wheels accelerate the contact patch back under the CoM and convert the impulse into a controlled excursion, whereas laterally no actuator produces ground force at all and the only restoring authority is the static w_{track} = 0.278 m track width plus hip-roll PD. We compare axis *means* rather than ranks because the raw ordering is contaminated by the second regularity. Second, the envelope is measurably *chiral*: 135° survives 40.7 N more than −135°, and −90° 28.3 N more than 90°. This is not sampling noise and not the plant, which is mirror-symmetric to numerical precision: it is the sign convention of the controller's antisymmetric hip-roll and hip-yaw channels, which leaves the closed loop C_2-equivariant but mirror-broken. The text traces it through the $\sin 2\theta$ harmonic and closes the argument with two control experiments. $F_{\min}$ (red) is the bearing-wise minimum and is the figure of merit quoted throughout the paper, because it is the magnitude survived in *every* direction; $F_{\max}$ (blue) is shown only to indicate the spread.

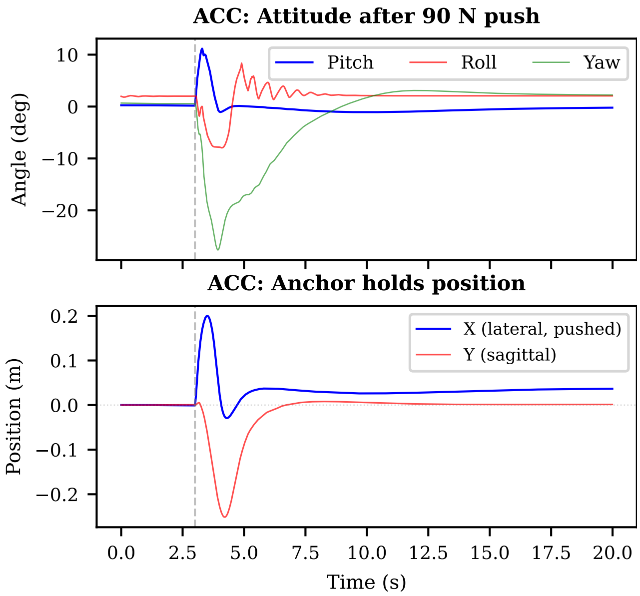

Figure 5: Post-push recovery (90 N lateral, world $+x$, at t=3 s). **Top:** attitude angles; stabilization within 2 s. **Bottom:** planar CoM position, which separates the two axes cleanly. The lateral push couples through roll into a large *sagittal* excursion (−251 mm peak); the anchor drives that axis back to 1.37 mm of home and holds it there at 0.018 mm SD over the final 2 s, a 99.5 % recovery. On the pushed lateral axis, which has no actuator authority (Section 5.1), the 200 mm peak excursion decays only to a permanent 36.2 mm offset, 18 % of the peak. The contrast between the two traces is the clearest single illustration of what the anchor does and where it does not act.

2 m straddle. A settling criterion alone cannot see this failure, because a robot that yaws off the side of the curb halfway along still lands on flat ground, settles and would score a pass on the tail criteria. Under the traversal criterion, and with the straddle gating of Section 4.6, roll on the one-wheel curb at 0.15 m/s measures 2.4±0.1° at 10 cm, 2.9±0.2° at 15 cm and 3.9±0.1° at 20 cm, with 10 of 10 traversals at every height as Fig. 6 shows. The residual growth with step height is tracking error rather than a geometry failure, since mid-straddle at 20 cm Eq. (22) commands a leg-length difference of 0.193 m against the 0.20 m step, leaving ε=7 mm unspanned and hence a predicted $\arctan(\varepsilon/w_{\text{track}})$=1.4° of geometric roll, so the remaining 2.5° is servo error arising because in this deeply flexed configuration the compressed leg carries its load at the shortest gravitational moment arm the PID gains were tuned for. The joint budget tightens with height without binding at 20 cm, the compressed leg's commanded knee angle growing from 2.18 rad to 2.58 rad against a 2.70 rad limit. With a split-independent roll band in the stability envelope, by contrast, heading hold is suppressed through the straddle, the mount transient is left undamped, the yaw error runs to the 0.30 rad command leash and the elevated wheel leaves the slab 1.3 m into the curb, whose tight standard deviation is a deterministic runaway rather than genuinely low run-to-run variability, the measured roll being 14.9±0.1° over ten trials with none traversing.

The two component ablations are unambiguous. Replacing the split with a symmetric posture that commands both legs to the same height causes a fall at every tested curb height without exception, the torso rolling past 60° within 0.8 s of curb entry, so the height split is load-bearing for this task in a way the flight override is not. Disabling the closed-loop leveling trim alone leaves the robot upright and traversing on the 10 cm curb at 4.5±2.3° but collapses the 15 cm curb to 1 of 10 and the 20 cm curb to none of ten with five outright falls, because the feedforward height-to-leg-length table alone does not span a step this large and the residual it leaves grows faster than the straddle allowance can absorb.

Driving off a ledge whose edge crosses the path at an angle ϕ releases the two wheels a stagger interval $w_{\text{track}} \tan\phi/v$ apart, and only within that interval can the leg split absorb the step. Instrumenting a 30° exit over a 20 cm step at 0.7 m/s shows the window closing in 0.18 s, the split having commanded 25 mm of differential leg travel against the 200 mm step when the trailing wheel lost contact so that the torso absorbed the remainder at −33.1° of roll against the $\arctan(0.20/w_{\text{track}})$=35.7° a wholly unspanned step predicts. The legs were not at their travel limit when contact was lost, with 99 mm of calibrated range unused, so the proximate failure is a race rather than a saturation, though relieving it would not rescue this case because 20 cm is itself unspannable at nominal stance and the ±6 cm leveling trim is unavailable throughout, its gate requiring both wheels loaded. Failure is consequently not monotone in step height, two regimes separating according to whether the leading wheel reaches the lower ground before the trailing wheel releases, since free fall over the step takes $\sqrt{2h/g}$ and is comparable to the stagger interval at these heights. A shallow step lands the leading wheel while the trailing wheel still bears on the platform, forcing the straddle the legs cannot span, whereas a deep step releases both wheels together and is handled by the flight path of Section 4.5, so widening the straddle regime requires knee range rather than gains.

We calibrate the boundary over 192 trials spanning 12 heights from 12 cm to 40 cm, eight independent initial conditions and two angles, with 43 of 96 settling at 30° and 24 of 96 at 45°. Both regimes are real and the dead band between them is wide. At 30° the shallow window runs to 15 cm at full success, degrades through 17 cm and 20 cm, and closes completely over 22 – 27 cm, after which the deep regime reopens from 30 cm and peaks at 37 cm. At 45° the shallow window is essentially gone, holding at 12 % for the shallowest step and at zero from 15 cm to 25 cm, and the deep regime opens later, at 27 cm. That the window shrinks with angle is what the stagger interval $w_{\text{track}} \tan\phi/v$ predicts, and it shrinks by roughly the right factor. Two features are worth stating as measured rather than smoothed, in that neither regime reaches full success at depth, the deep-regime maximum being 75 %, and the 45° sequence has a reproducible empty cell at 32 cm sitting between 50 % and 62 % neighbours that the two-regime account does not predict. The boundary is therefore calibrated as a success-probability field over launch height rather than as a single threshold.[6]

The gate structure isolates each extension to its physical regime without re-tuning the anchor, and both extensions cost three lines of controller code each. Their measured necessity nevertheless differs sharply, since the per-leg height split is required for the curb task at every tested height whereas the flight override is a variance-bounding refinement that only pays at the top of the drop envelope.

One-wheel curb (20 cm) — per-leg terrain adaptation

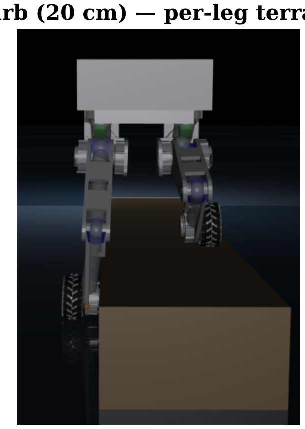

Figure 6: One-wheel curb (20 cm). With the per-leg height split and the straddle gating of Section 4.6: 3.9±0.1° straddle roll, 10/10 traversed. With a symmetric posture (split removed): the curb-side leg cannot fold far enough, roll diverges past 60° within 0.8 s of curb entry, 0/10.

5.4. Comparison with Classical Baselines and a Preliminary RL Reference

No balance controller has been published for this morphology, a 10-DOF plant carrying three hip degrees of freedom per leg, so there is no established classical design for it to compare against. Whleaper [15] is the closest published platform, and it switches between an LQR and a separately trained policy by locomotion mode rather than closing every axis with one controller. What can be measured in the absence of such a design is transfer, meaning how far the reduced-order controllers the wheeled-biped literature does publish carry when applied to this plant. The rows below should therefore be read as a transfer result rather than as a ranking of ACC against optimal control, and the row that tests the linear-design family on its own terms is the full-state controller of Section 5.4.2, designed on a linearization of this plant rather than of a simpler one.

Six classical baselines are evaluated alongside a preliminary model-free PPO run, all sharing identical environment setups. Four are 4-state TWIP variants [48, 49], namely a plain LQR with decoupled roll and yaw PD through a PID servo layer of roughly 0.25 s lag, the same design augmented with a pitch-error integral under conditional anti-windup clamped to ±3.0 rad/s, a direct-torque LQR re-optimized for the zero-lag plant, and a PI variant that adds integral anti-windup on forward position behind a ±15 cm conditional gate. Two more are coupled 6-state designs that extend the 4-state model with the roll states $(\phi, \dot{\phi})$, one acting through the servo layer and one emitting direct torques with gains re-derived for the zero-lag plant and leg posture held by PD position control. The seventh entry is a model-free policy trained from scratch for 5.4M environment steps in `BalanceEnv` on a single seed without domain randomization or hyperparameter search, included as a difficulty probe rather than as a tuned RL baseline in the sense of Limitation 4. Direct-torque LQR removes the servo path and with it both the PID gains and the measured actuation lag, while the PI variant adds the standard engineering fix for a DC position bias together with windup protection, its anti-windup logic verified independently by `scripts/validate_lqr_aw.py`. Every gain is listed in the preceding paragraph, given in full because the central claim of this subsection is a negative one, that no member of this family stands, and a negative result is only as strong as the tuning behind it.

Every episode ends on roll. LQR balances wheeled bipeds on real hardware, as both Ascento [1] and DIABLO [14] demonstrate, and the axis that terminates these episodes is not the one those platforms exercise. None of these controllers fails at the task LQR is credited with there, since at 100 Hz the servo variants hold pitch to 0.006° RMS and the full-state design of Section 5.4.2 holds 0.12° for the full 20 s. What terminates every episode is roll, at 24.2° RMS for the servo variants and 21.2° for direct torque, on a morphology carrying two lateral degrees of freedom per leg that those platforms do not have. The published designs do not ask a single LQR to close that axis either, and DIABLO runs separate yaw, split-angle, height and roll loops around its balance controller. Here the 4-state model carries no lateral state at all, while the 6-state model carries roll but linearizes it about one operating point and then commands the hip-roll channel against its ±0.7 rad joint limit for the whole episode at every gain in an eightyfold sweep. These rows therefore support a narrow morphological result, that reduced-order LQR of the form the wheeled-biped literature uses does not transfer to a 10-DOF articulated-leg plant as a single controller for all axes. Whether LQR can balance a wheeled biped at all is settled elsewhere and in the affirmative, and Section 5.4.2 removes our tuning from the statement by designing on the plant itself.

The same result appears inside ACC's own ablation ladder from the other direction. Rung L0 is proportional-only on the wheel channel, carrying no position term, no integral and no anchor, yet it stands in all ten trials where an optimal law does not. L0 is not a linear controller for the whole robot but ACC with the anchor removed, and it retains the dedicated roll, yaw and posture channels that no row of

[6] `scripts/ramp_step_tests.py` with `--seeds`; raw output `outputs/oblique_regime/diag30_fine.txt` and `diag45_fine.txt`. Each seed perturbs the initial joint angles by 3 mrad and the root by 1.5 mm, so the eight trials per cell sample initial conditions on an otherwise deterministic course.

Table 4 has. The comparison in that table is therefore neither optimal against hand-tuned nor linear against nonlinear but one linearization for all axes against per-axis structure. Read that way, the ladder and the baseline table state the same result from opposite directions. Remove the structure and add optimality and the plant falls in 0.32 s, whereas keeping the structure and stripping everything else back to a proportional term leaves it standing at 17.461 mm repeatability inside a 69.7 mm limit cycle. That gap between L0 and ACC is what the remaining terms buy, and it is measured in Section 5.1 rather than asserted here.

Every baseline weight follows Bryson's rule [50], with $Q_{ii}=1/x_{i,\max}^2$ and $R=1/u_{\max}^2$, and a 46-configuration retuning sweep confirmed that no retuning within these ranges changes the outcome. The 4-state model carries pitch, pitch rate, forward velocity and forward position and is solved at $Q=\mathrm{diag}(10,2,3,0.3)$ with R=0.8, roll and yaw being closed by decoupled PD through the PID servo layer, while the anti-windup variant adds a pitch-error integral with conditional anti-windup clamped to ±3.0 rad/s. The 6-state model adds roll and roll rate and is derived in continuous time at $Q=\mathrm{diag}(10,2,3,0.5,3,0.3)$ with $R=\mathrm{diag}(0.8,1.0)$, shared by its servo and direct-torque variants. Direct torque here means the PID servo path removed and the servo lag τ_s set to zero, which requires re-optimization, giving $k_{p,\mathrm{roll}}$=55.0 in the 4-state case and K_{pitch}=113 with K_{roll}=58 Nm/rad in the 6-state case, and the PI variant adds an integral on forward position at $k_{i,\mathrm{pos}}$=8.0 rad/s/(m s) clamped to ±5.0 rad/s behind a ±15 cm conditional gate. The servo layer itself runs $k_p=[55,40,70,70,4]$, $k_d=[3,2,4,4,0]$ and $k_i=[0.8,0.4,1,1,0.1]$ ordered hip roll, hip yaw, hip pitch, knee and wheel, and its lag of 0.25 s is measured as the wheel-velocity 90 % rise time, roughly 25 steps at 100 Hz.

Table 4 reports the comparison, and all six classical baselines fall within 0.60 s. Removing the PID servo path cuts torque by a factor of 2.5 and nearly doubles survival from 0.32 s to 0.60 s, yet it worsens roll from 24.2° to 27.5° RMS, because the recovered authority is spent in the sagittal plane the model describes rather than in the lateral one that ends the episode. Adding a forward-position integral gives back 0.08 s of that gain rather than adding to it, since the integral cannot accumulate usefully before the roll-induced fall, which confirms that the 4-state TWIP model is structurally insufficient for this articulated-leg platform. The preliminary PPO reference sustains single-height standing longer at 3.20 s but exhibits an 85 % fall rate across multi-height command transitions with 54.2 mm height error, and that is the observation the structured prior is meant to address, since unaided search spends its budget rediscovering the pitch–height coupling that the posture map and the gate schedule encode directly. ACC by comparison achieves 0.060±0.006° pitch RMS, 0.010±0.001° roll RMS, 0.18±0.02 mm CoM height RMS and 3.28±0.00 Nm torque RMS over ten episodes, and the 46-configuration sweep along the dimensions of the preceding paragraph found no exception anywhere among the tested variants.

Table 4
Classical baselines and the preliminary PPO reference vs. ACC, clean sensors. **Read the two attitude columns together:** every classical row is terminated by *roll*, not pitch; θ_{RMS} stays between 0.001° and 0.181° while ϕ_{RMS} runs 21 − 28°. None of these controllers fails at the sagittal task LQR is credited with on Ascento and DIABLO; they fail on a lateral axis their model does not carry. All rows at 100 Hz control except the PPO reference, which is reported at the 50 Hz rate its policy was trained at. Parenthesised N is the episode count. See the note below the table.

Configuration	Surv. (s)	θ_{RMS} (°)	ϕ_{RMS} (°)	H_{RMS} (mm)	τ_{RMS} (Nm)
LQR, TWIP (N=20)	0.32	0.006	24.2	205.8	21.99
LQR + AW (N=20)	0.32	0.006	24.2	205.8	21.99
LQR, DT (re-opt.) (N=20)	0.60	0.033	27.5	119.2	8.76
6-St. LQR (N=20)	0.32	0.006	24.2	205.8	21.99
PI + AW, DT (N=20)	0.52	0.181	26.1	171.3	10.42
6-St. LQR, DT (N=20)	0.16	0.001	21.2	128.5	14.44
PPO ref. (50 Hz) (N=20)	3.20	4.82	3.15	54.2	18.40
ACC (N=10)	**20.00**	**0.060**	**0.010**	**0.18**	**3.28**

LQR / LQR+AW / 6-St. use the PID servo; LQR (DT), PI+AW (DT) and 6-St. (DT) command torque directly; PPO is a 5.4M-step single-seed run without domain randomization, a reference point rather than a tuned RL baseline. The three identical rows arise because all command through the same PID servo path, whose response dominates before their gain differential can act, a coincidence that holds at both control rates (Section 5.4.1). Between-episode standard deviations (omitted above for legibility): survival time is deterministic on all six classical rows (±0.00 s at N=20, seeds $\{0,42,123\}$); PPO: ±0.85, ±0.91, ±0.45, ±8.3, ±2.1; ACC: ±0.00, ±0.006, ±0.001, ±0.02, ±0.00. Per-metric between-episode dispersions for the classical rows are not retained in the rate-matched output (`outputs/rate_matched_baselines/results.json`), which stores episode-aggregated values.

N=20 throughout; classical rows from `outputs/rate_matched_baselines/results.json`.

5.4.1. *Both control rates give the same outcome*

Table 4 reports every classical baseline at ACC's 100 Hz, so the headline comparison carries no control-bandwidth differential, but those gains were tuned for a 50 Hz loop and all six were therefore re-run at both rates. Doubling the control rate rescues none of them. The fall rate is 100 % in all twelve cells and survival time improves for no variant, shortening for five of six. The diagnostic quantity is roll, which sits at 24.21° at both rates for the three servo variants and rises from 21.22° to 27.48° for the direct-torque LQR, while faster sampling improves the quantities the controller actually regulates, pitch RMS falling roughly sixfold for the servo variants and torque RMS dropping from 92.1 Nm to 22.0 Nm. Leaving the lateral divergence that terminates the episode untouched is the signature of a model-structure deficit rather than of a bandwidth deficit, since the 4-state TWIP model carries no lateral state at all and the 6-state variant linearizes roll about a single operating point. Reporting the baselines at 100 Hz is thus the rate-matched choice and also the harsher of the two for them, since five of the six survive longer at 50 Hz.[7]

5.4.2. *A full-state LQR designed on the plant itself*

All six baselines above are designed on a reduced model, which confounds the method with the model it is designed on. The control experiment that separates the two is a linear-quadratic regulator derived from the plant directly. It begins from a damped Gauss–Newton solve for a true static equilibrium $(\mathbf{x}^*, \mathbf{u}^*)$ at the commanded height, takes central differences on the whole held-input control step of the complete 16-DOF, 10-actuator MuJoCo model for the transition Jacobians, removes the two cyclic wheel-angle states and closes with a discrete-time Riccati solve on the remaining 30. Torque is emitted under the same 400 Nm/s rate limit as every other row, and Bryson's rule [50] fixes $\mathbf{Q}$ and $\mathbf{R}$ from physical maximum deviations, leaving one knob, a uniform control weight r, swept over six decades rather than tuned. The design is sound on every diagnostic that bears on it, since at the 0.65 m equilibrium the residual unbalanced force is 0.428 N against 79.46 N of body weight, the reduced $\mathbf{A}$ has full rank and is PBH-stabilizable, the Riccati relative residual is of order 10^{-14} and the designed closed loop has max $|\lambda|$=0.9911. The one figure that invites a disqualifying reading, $\kappa(\mathbf{A}) \approx 10^{10}$, does not support it, being a 50 Hz artifact that falls to 2.6×10^7 at 100 Hz and tracing to a smallest singular value of order 10^{-9} belonging to strongly contracting contact modes. Whatever limits this controller, it is not the linearization.

What the sweep shows instead is a slow divergence that a short horizon hides. Over 20 s the design looks successful, r=1000 giving no falls at h=0.65 m with 11.0 mm height RMSE, yet at 60 s every configuration tested falls in every episode between 22.2 s and 57.0 s, with planar drift growing monotonically to 0.43 – 0.63 m, and that drift rather than attitude is what ends them. Nor does one weight cover the band, since r=100 stands 20 s at 0.60 m and 0.69 m but falls at 0.65 m while r=1000 does the reverse, under uniformly random height commands the fall rate runs 45 – 80 % across the three usable weights, and at the baselines' own 50 Hz rate every weight falls inside 20 s. Push separates it from ACC outright, with total falls under the standard 50 N impulse and a maximum recoverable push of 10.0 – 20.2 N against ACC's 82 N. The reading is therefore not that the plant defeats linearization but that full-state optimal linear feedback about a single equilibrium buys roughly two orders of magnitude of survival over the reduced-order designs and still does not close the problem, diverging slowly rather than quickly, holding one height at a time rather than the band, and carrying essentially no push envelope. Two of those three are exactly what the anchor integral and the gate schedule are for, which is why the design is reported as a control experiment rather than promoted into Table 4, since at the 20 s horizon used for every other row it would read as a partial success. That argument only carries if the proposed controller is held to the same test, and it was, extending the quiet-stance window by the same factors returning the opposite verdict for ACC, which settles in every trial at 300 s and 1800 s with residual drift of 0.0004 mm/min.[8]

5.4.3. *Sensitivity of the coupled model's empirical roll coupling*

The two 6-state coupled baselines close their roll channel with a constant $b_{r,h}$ mapping hip-roll angle to roll acceleration that the planar pendulum model does not supply in closed form, and it was set empirically to −5.0. Since the negative result rests on those rows, that constant was swept over nine values from −0.5 to −40 with gains re-derived through the same Riccati solve at each value, and the outcome does not move, the servo variant falling at 0.320 s in every episode for every value and the direct-torque variant at 0.160 s, roll RMS staying constant to three decimals in both. This is invariance to the parameter rather than insensitivity of the metric, because the re-derived roll gain varies thirtytwofold across the sweep for the servo path. The mechanism is saturation, since the roll-channel command reaches the ±0.7 rad hip-roll joint limit at a roll excursion of 0.55 – 17.87° depending on $b_{r,h}$ while the excursion observed before the fall has an RMS of 24.2°, so the channel is commanded against its limit throughout the episode and the gain that computed the command cannot matter.[9]

5.5. Whole-Body Control Baselines

To address whether a tuned whole-body controller could match ACC we evaluated two configurations developed and subsequently discarded during controller development. The first is a task-priority quadratic-program WBC minimizing CoM height error, torso orientation error, posture damping and wheel regularization subject to full-body dynamics and contact constraints, used as the primary balance controller. The second blends the same WBC as an assist on top of ACC through $\boldsymbol{\tau}_{\text{assist}} = \boldsymbol{\tau}_{\text{ACC}} + \alpha(\boldsymbol{\tau}_{\text{wbc}} - \boldsymbol{\tau}_{\text{ACC}})$ and is evaluated both with the per-joint authority gate active and with it disabled. Both run under identical cloned conditions as a three-arm counterfactual over 225 scenarios spanning fixed-height balance, height transitions and push recovery from 20 N to 120 N, totalling 481 000 control steps with no gain retuning, and so that the comparison is not a verdict on our own implementation the WBC was verified beforehand against executable references as Appendix B records.

[7] N=20 episodes per cell, seeds $\{0, 42, 123\}$, clean sensors, zero actuator delay, episode length fixed in seconds rather than in steps. Produced by `scripts/collect_rate_matched_baselines.py`, which adds a `-control-hz` flag to `scripts/eval_balance.py`; raw output `outputs/rate_matched_baselines/results.json`. The LQR feedback gains are continuous-time and therefore rate-independent, so the rate change affects only the substep count, the drift and integral accumulators, the torque rate limiter and, for the finite-difference-linearized variant, the linearization step.

[8] N=20 episodes × seeds $\{0, 42, 123\}$ per configuration at 100 Hz; between-episode SD is 0.00 s on time-to-fall, and weights r<100 fall within 1.4 s. Produced by `scripts/eval_balance.py--controllerbaseline_full_lqr`; per-configuration results and the design diagnostics are in `outputs/fullstate_lqr_corrected/`, whose `README.md` maps each row to its run.

[9] Twenty episodes over three seeds at 100 Hz, both variants; raw output `outputs/b_roll_hip_sweep/results_100hz.json`, produced by `scripts/sweep_b_roll_hip.py`.

Used as the primary controller, the standalone QP-WBC falls in all 225 scenarios where ACC falls in none, holding the robot upright for 0.570 s and spending 97.3 % of every episode in a fallen state. The failure is lateral, roll RMS reaching 93.45° against ACC's 3.71° and peaking past horizontal at 103.9° while pitch RMS reaches only 2.33°, and base height collapses to 0.091 m. The fallen fraction is uniform across all five scenario suites, so the collapse is specific to neither pushes nor commanded height changes.

The failure is invariant to implementation quality. A degraded build of the same controller whose QP returns a usable solution on only 85.85 % of steps falls at 0.52 s while the verified build, solving 99.99 % of steps, falls at 0.570 s, so a fourteen-point improvement in solve rate buys 50 ms of upright time against a 0.570 s lifetime. Nor is it an artifact of one task weighting, since re-running the WBC-only arm at four weightings over an identical scenario set moves the fallen fraction by 0.1 percentage points and upright time by 0.03 s, the best mode improving pitch RMS by 40 % at the cost of roll because a single SO(3) orientation task carrying equal roll and pitch gains trades one axis against the other rather than fixing both.

The mechanism is that the WBC's posture task is a velocity damper rather than a position tracker. Its reference is rebuilt from the currently measured joint vector at every control step, so the posture error vanishes identically and the task collapses to $-k_{d,\text{posture}}\dot{\mathbf{q}}$ with $k_{d,\text{posture}}$=2.0, which can slow a postural drift but cannot reverse one. Combined with the absence of gravity-compensation feedforward, nothing in the task stack restores the legs once a lateral excursion begins, which explains why the collapse is lateral, why it is insensitive to the task weighting and why solver quality does not delay it. The gap is one of structural expressiveness rather than of torque optimality, because a WBC distributes torque optimally for a fixed task set but cannot express quiet stance, push ringdown and large excursion as distinct control modes, and the precision–recovery trade-off ACC resolves through spatial gating and asymmetric envelope-following is not expressible as a task-priority QP objective. Even an ideally implemented WBC would therefore lack the prior that gates the anchor integral and would face the global-integral windup that ablation S4 confirms is fatal for sub-millimetre precision.

We ran the assist in both of its available configurations, which together constitute the gate ablation reported in Table 5. With the gate disabled, so that α is fixed at 0.25 and per-joint authority is capped at 20 % of the ACC torque, WBC torque entered the blend unconditionally. That arm never falls because ACC carries it, yet it degrades ACC on every metric except one, raising pitch RMS from 0.520° to 1.295°, yaw drift from 6.69° to 12.56°, planar drift from 0.099 m to 0.199 m and torque RMS from 3.18 Nm to 3.67 Nm. With the gate enabled, so that per-joint authority becomes $\alpha_j = \alpha_{\max} g_{\text{stability}} g_{\text{height}} g_{\text{push}} g_{\text{divergence}} A_j K_{\text{role},j}$, instrumenting every factor at every control step shows the admitted authority averaging 0.0061 on the one height variant where the gate opens at all, some 2.5 % of the permitted maximum, while on the remaining four the height factor falls below its 1×10^{-3} cutoff on the commanded height alone and α is identically zero, so the gated assist is indistinguishable from ACC alone to four significant figures. The two configurations therefore bracket the assist, since where WBC authority is admitted it degrades ACC and where it is harmless it is inactive, and no setting of the gate makes it simultaneously active and beneficial.

Table 5
Three-arm counterfactual under cloned initial conditions over 225 scenarios and 481 000 control steps, comparing ACC alone, the standalone QP-WBC, and the WBC blended as an assist on ACC with its authority gate disabled. All arms use the verified WBC implementation of Appendix B.

Metric	ACC	WBC	Assist
Scenarios with a fall	0/225	225/225	0/225
Episode fallen (%)	0.0	97.3	0.0
Upright time (s)	full	0.570	full
Pitch RMS (°)	0.52	2.33	1.295
Roll RMS (°)	3.71	93.45	**3.20**
Roll max (°)	8.37	103.89	7.64
Yaw drift (°)	6.69	6.80	12.56
Base height RMS (m)	0.540	0.126	0.518
Final height (m)	0.538	0.091	0.515
Planar drift (m)	0.099	0.165	0.199
Torque RMS (Nm)	3.18	0.96	3.67
QP success (%)	—	99.99	100.00

"Full" upright time denotes no fall in any episode. Base height RMS is the root-mean-square of *absolute* base-z, so a larger value means the robot stayed up and WBC's 0.126 m is a collapsed torso rather than tighter tracking, just as its 0.96 Nm is the torque of a machine lying on the ground. Roll is the sole metric on which the assist beats ACC (bold). The gated column is omitted because on the four height variants where its gate is identically closed it was not run, and on the fifth it agrees with the ACC column to four significant figures.

Roll is the one metric WBC improves, and it is the exception in both configurations, improving from ACC's 3.71° to 3.20° over the full batch and from 3.55° to 2.90° on the matched subset, which makes it systematic rather than sampling noise. ACC regulates the lateral axis with decentralized hip-roll PD whereas the WBC uses a centralized orientation task exploiting full mass-matrix coupling, so on that axis alone the centralized formulation is the better regulator, but buying 18 % of roll costs two and a half times the pitch error and twice the drift. A centralized lateral task is therefore a credible direction for improving ACC's weakest axis, the 98.0 N mean lateral push bearing of Section 5.2, though not through torque-level blending with a posture stack that cannot hold a posture. Timing is not what excludes it, since the QP solve costs 0.169 ms mean and 0.30 ms at the 99th percentile over 481 000 solves with OSQP [51], comfortably inside the 10 ms control period, so the assist is excluded on control grounds alone.

5.6. Architectural Analysis

We evaluate ACC along four dimensions, namely robustness to noise and delay, gate parameter sensitivity, torque rate-limit invariance and compound-disturbance recovery.

A factorial sweep over four noise levels and three delay levels, with twenty trials in each of the twelve cells, bounds the first two axes at once.[10] Measured cell by cell, $F_{\max}$ is 95.7±0.8 N on clean sensing at zero delay, 96.0±1.0 N under low noise and 93.5±5.2 N under medium, falling to 50.5 – 51.3 N across all three at 10 ms, while idle lateral RMS runs 0.43 mm, 5.86 mm and 30.82 mm undelayed and 2.43 mm, 10.66 mm and 35.38 mm at 10 ms. One control period of delay proves benign for stability and expensive for performance, since at 10 ms there are no falls on clean and low noise but $F_{\max}$ bifurcates from 95.7 N to 50.5 N and idle RMS rises more than fivefold. Three control periods is a different regime entirely, the platform falling in every trial at 30 ms on clean sensing and equally under low and medium noise with no recoverable push at any tested magnitude, so delay tolerance is bounded rather than open-ended and the bound lies between one and three control periods, which the finer sweep below locates to 14 – 16 ms. Noise alone is survivable up to the medium level at zero delay. What ACC does not tolerate is the conjunction, since medium noise together with 10 ms of delay costs two falls in twenty trials, the only sub-catastrophic falls anywhere in the grid, every other fall being a cell in which all twenty trials fail, while high noise is unrecoverable at every delay including zero because the asymmetric envelope cannot distinguish noise from motion, which forces anchor disengagement and leaves a residual proportional mode that cannot hold the platform. That conjunction is the binding constraint on the design, and Section C shows it is what prices the one gain change that would otherwise remove ACC's parking offset.

That seventyfold idle degradation from clean sensing to medium noise at zero delay is the sharpest single sensitivity in the design, produced by the same mechanism, since the envelope gate cannot separate sensor noise from real motion and so releases the integral and re-engages it repeatedly. That locates the boundary of the anchored regime, because below it the anchor delivers the precision of Table 2 whereas at medium noise it no longer pays for itself, the un-anchored L0 baseline holding 3.675 mm on the same axis. The design requirement that follows is explicit and testable, namely that ACC is specified against a filtered velocity estimate, which is a requirement on the state estimator rather than a limit on the controller. One convention governs these figures, namely that the idle column is lateral rather than sagittal and uses a 3 s settling allowance, so its clean reference is the 0.43 mm entry and not ACC's lateral 0.730 mm. Every cell is measured identically and the degradation factors are the quantity the sweep establishes.

[10]Noise at 1σ: low is 0.01 rad/s on angular velocity, 0.01 m/s^2 on gravity and 0.001 rad on joint position, medium is 5× and high is 10× that. Delay is a transport lag on the actuator command, the torque written n control periods ago being the one the plant receives. Idle RMS is the lateral CoM standard deviation over 20 s after a 3 s settle from EMA_0=0, averaged over surviving trials only, so the medium-noise 10 ms cell is mildly optimistic and its falls column is the primary metric; $F_{\max}$ is a binary-search lateral push to ±5 N with a 10 N search floor. Produced by `scripts/collect_robustness_sweep.py`, raw output `outputs/paper_statistics/robustness_sweep.json`.

The grid above quantises delay to the control period, so it brackets the useful range at 10 ms and 30 ms without saying where inside that bracket the behaviour changes. The knee was therefore located separately with the transport delay applied per physics substep, which resolves the axis five times more finely and is the more faithful model, since a real actuator's latency is not quantised to the controller's period. Clean sensing makes the harness deterministic, every multi-seed cell having returned identical values across seeds, so single-seed cells are exact rather than sampled. That sweep places the knee between 6 ms and 8 ms, $F_{\max}$ holding 95.5 N, 94.4 N and 93.2 N out to 6 ms and then collapsing by 45 % to 51.0 N, thereafter declining monotonically through 41.6 N at 12 ms and 28.8 N at 16 ms to the search floor by 20 ms.[11] That tail distinguishes a genuine loss of margin from a narrow phase-alignment notch, and it is the former. The transition is a cliff rather than a roll-off, so a design either clears it or does not, and the knee sits below the 10 ms figure the coarse grid implied while the end-to-end budget estimated in Section 6 lands inside the 6 – 8 ms bracket rather than clear of it, which makes achieved latency rather than architecture the deciding quantity. Absolute levels are not directly comparable with the coarse grid, whose push protocol differs, but the two harnesses agree on both anchors at 95.5 N against 95.7 N undelayed and 56.9 N against 50.5 N at 10 ms.

Losing the push margin and losing the platform are different thresholds, since past 8 ms the robot still stands and simply stops absorbing pushes. The delay at which it stops standing at all was measured with the same substep operator on a quiet-stance trial, and ACC holds station through 14 ms though visibly degraded, its idle lateral RMS running 2.47 mm at 10 ms and 10.84 mm at 14 ms against 0.43 mm undelayed, falling at 16 ms and beyond. The threshold is thus 14 – 16 ms, bracketed to one physics substep, and above it the collapse is prompt at 0.7 s for 30 ms, which is what makes the 30 ms column of the coarse grid unanimous. One consequence bears on how the 16 ms figure should be read, because at that delay the quiet-stance trial diverges only at 22.0 s, later than the push harness's 20.1 s horizon, so the 28.8 N recorded there is measured on a platform that is already unstable and merely slow about it, and the last delay at which $F_{\max}$ describes a platform that also survives the full quiet-stance window is 14 ms.

Whether the knee belongs to the architecture or to the shipped gain set is a question the sweep answers directly. Holding the delay at 8 ms and sweeping one profile constant at a time gives not a gently sloped optimum but a sharp local dip, since the shipped value of each of the four constants tested scores 51.0 N while nearly every neighbouring

[11]Transport delay applied per physics substep, 2 ms at 500 Hz, which bounds the resolution and is why neither knee is localized further; $F_{\max}$ by binary-search lateral push over seven iterations across 10 – 160 N, clean sensing, N up to 5. The 10 ms cell is the only pure one-control-period hold, the others mixing two consecutive commands within a period, which is why $F_{\max}$ is not smooth across it. Produced by `scripts/delay_cliff_resolution.py` and `scripts/delay_retune_sweep.py`; raw output in `outputs/robustness_sweep/delay_cliff/` and `.../delay_retune/`.

value, in either direction and on any of the four, recovers 78 – 93 N.[12] A margin that is genuinely exhausted does not behave that way, and the shape is the signature of an unlucky phase alignment at one operating point, which makes the 6 – 8 ms knee a tuning artefact rather than an architectural ceiling.

Carrying the largest single-knob change forward, namely doubling the scheduled sagittal damping from 22.5 to 45 Nm/(m/s) to give the variant we denote ACC-DH, gives a controller identical to the shipped one at the shortest delays, no worse at any delay tested, and with the knee moved outward by one grid step to 8 – 10 ms, holding 93.2 N at 8 ms where the shipped gains give 51.0 N and adding 44 – 90 % of margin across 12 – 16 ms, though both arms reach the search floor by 20 ms, so the retune buys headroom below the cliff without relocating the eventual failure. The measured price is negligible at zero delay, undelayed idle lateral RMS rising only from 0.790 mm to 0.801 mm, and real under delay, ACC-DH being the less steady of the two at 2.89 mm against 2.47 mm at 10 ms while falling at the same 16 ms, so the retune moves the push cliff without moving the standing threshold. Under the noise-plus-delay conjunction it is better rather than worse, its 8 ms $F_{\max}$ rising from 76.8 N to 92.0 N under low noise and from 84.3 N to 90.2 N under medium, which also shows that the shipped controller's 8 ms dip is deeper on clean sensing than under noise and that the clean-sensing cliff is the conservative end of the range. We report ACC-DH as an evaluation variant and do not adopt it, since the constant it changes also governs driving acceleration and settling and its cost on the driving, terrain and flight axes has not been re-measured, a 48-scenario qualification suite rather than a single-axis sweep being what decides a default. The narrower claim is sufficient for the limitation it addresses, that the delay knee is movable by tuning at negligible cost to undelayed quiet stance, but that it buys push margin only and leaves the 14 – 16 ms standing threshold where it was.

Finite-difference gradients over the gate parameters reveal a stark hierarchy in which the release coefficient α_r dominates at a sensitivity of 220.4, twenty times the next parameter and a hundred and thirty times the pitch gates, whose near-zero sensitivity marks them as safety interlocks rather than tuning knobs. Porting to a new morphology therefore starts by identifying the dominant ringdown period from post-push decay and setting τ_{release} to three to five times it, then setting the quiet-stance band at two to three times the idle velocity noise floor and the pitch gates conservatively at $\pm 12°$ and ± 15 °/s, a fixed k_p=50 and anchor integral gain having sufficed here. The gate failure modes themselves map onto measured ablation rows, which gives implicit FMEA coverage, since a gate stuck open causes the windup and fall of S4, a gate stuck closed yields the proportional-only limit cycle of L0, and a gate stuck half-open triggers the parametric resonance of S5.

The 400 Nm/s slew limit used throughout is not what sets ACC's push ceiling. Sweeping it over an eightfold span, with each trial drawing an independent initial-posture perturbation from the same distribution as the push protocol of Section 5.2, gives lateral $F_{\max}$ of 92.5±1.1 N at 100 Nm/s and 94.4±0.0 N at 200, 400 and 800 Nm/s, with idle lateral RMS between 0.695 and 0.738 mm across the four settings, a 2 % spread in push margin and a statistically indistinguishable spread in idle precision. The limiter is genuinely active where it could matter, since instrumenting the commanded torque during a 90 N push shows the slew saturating exactly at each configured bound and even at 800 Nm/s the wheel channel still reaches it, whereas during quiet stance the peak commanded slew is 0.51 – 1.66 Nm/s, some sixty to a hundred and ninety times below even the 100 Nm/s bound, so idle precision cannot be slew-limited at any tested setting. The ceiling is therefore architectural rather than actuator-limited.[13]

The push protocol of Section 5.2 holds the height command fixed, but because the posture map is height-scheduled a commanded transition and a push compete for the same leg-torque budget, so the two disturbances are evaluated together here. The scenario applies the standard lateral push impulse at the midpoint of a 1 s commanded height ramp of size Δh about the 0.404 m nominal CoM height and locates $F_{\max}$ by the same binary search, with Δh=0 serving as the matched static control so that the two differ only in the height command, ten trials to a row and each drawing an independent initial posture. A commanded height change must move both the gain schedule and the leg posture target, since ramping the schedule alone would leave the robot standing at its nominal height with detuned gains, so both are ramped and the achieved CoM height is verified against the model's own forward kinematics before each sweep.[14]

Three findings follow. The first is that inside the calibrated posture band a simultaneous height command is nearly free. Against the static control's 94.8±0.8 N, a +5 cm rise costs 5.1 % at 89.9±0.9 N and a −5 cm squat is better than static at 101.0±1.0 N, or +6.5 %, because squatting lowers the CoM and shortens the toppling moment arm while

[12] Velocity damping scale swept over {0.5, 0.75, 1.0, **1.5**, 2.0, 3.0} for 39.3, 42.8, 85.0, 51.0, 88.5 and 93.2 N; drift k_{vel} over {5, 10, **15**, 25, 40} for 65.1, 92.0, 51.0, 90.9 and 80.3 N; soft anchor k_p pitch over {20, 25, 30, **35**, 45, 50} for 85.0, 81.5, 85.0, 51.0, 78.0 and 86.2 N; anchor k_{vel} boost scale over {0, 2.5, **5**, 10} for 85.0, 92.0, 51.0 and 62.7 N. Shipped values in bold, all other constants held at their promoted values, same push protocol and bisection bracket as the resolution sweep, clean sensing and N=1.

[13] Sweep: `scripts/sweep_rate_limit_corrected.py`, raw output `outputs/rate_limit_sweep/results_corrected.json`; slew instrumentation: `scripts/probe_torque_slew.py`.

[14] Binary search over [10, 130] N; rows within ±5 cm stay inside the calibrated posture band 0.354 – 0.454 m while the ±10 cm rows are extrapolated. Against the static control every row is significant under Welch's t-test at the Bonferroni-adjusted α=0.0056 of Section 6 except −10 cm (p=0.99), and in the re-calibrated arm only the +10 cm row differs from the shipped map (p=1.5 × 10^{-4}), the other three lying within noise as they must, since the two maps agree to 3.2 mm inside the calibrated band. Peak pitch is measured over the whole episode at each trial's own $F_{\max}$, so it is a shape statistic at the survival boundary rather than a severity ranking across rows. Produced by `scripts/compound_disturbance_corrected.py`; raw outputs `outputs/compound_disturbance/results_corrected_map_shipped.json` and `..._map_comcalib.json`, with posture-map accuracy verified by `--self-check` against the model's own forward kinematics.

the push arrives. Both effects are statistically resolvable but operationally small, and since they bracket the static value there is no shared-budget penalty at this amplitude.

The second finding is that the degradation is one-sided, so that descent is free at any amplitude tested. Squatting -10 cm, fully outside the calibrated band, costs nothing measurable at 94.8±1.9 N with p=0.99, while rising $+10$ cm falls to 29.7±8.7 N, a loss of 68.7 %. Extrapolation as such is therefore not the mechanism, direction is. Rising drives the knee toward extension, where the leg Jacobian is worst conditioned and the same joint torque produces the least vertical force, and it does so while raising the CoM, whereas squatting moves in the opposite direction on both counts and pays for it only in peak pitch, the 28.9±1.8° of the -10 cm row being the largest measured, rather than in margin. A control experiment separates how much of the remaining collapse is calibration error and how much is the physics of standing tall. The shipped posture map is exact inside its band but drifts outside it, undershooting the commanded CoM height by 10.0 mm at 0.504 m and 29.1 mm at 0.554 m, and re-parameterizing the same posture family so that the achieved height matches the command, which changes no controller and is therefore only a relabelling, raises $F_{\max}$ at $+10$ cm by 78 % to 52.8±12.1 N with p=1.5×10^{-4} and drops the peak pitch from 12.6°±13.1 to 8.8°±2.3. It does not restore the static value, closing only 35 % of the 65.1 N deficit, so the calibration error is a real and removable contributor but a minority one and the residue is the genuine cost of carrying the load on a more extended leg. Inside the band the two maps are indistinguishable, which is the control the arm needs to be interpretable, and the large peak-pitch spread of the shipped $+10$ cm row, with one trial reaching 46.7°, is consistent with trials sitting on the edge of the capture basin, that spread rather than the mean being what the re-calibration removes.

The third finding comes from a scenario that tests sequential rather than simultaneous disturbance, a 90 N lateral impulse followed 2 s later by a 60 N counter-directed impulse delivered before the first ringdown has fully decayed. ACC survives **10/10** with a Clopper–Pearson 95% interval of [0.69, 1.00] and an episode peak pitch of $11.89 \pm 0.51°$, essentially all of which comes from the first impulse, since after the reversal the pitch peaks at only $2.78 \pm 0.11°$ and never re-enters the 5° band used as the settle criterion in any trial. The reversal is thus absorbed within the residual motion of the first recovery, which is exactly the behavior the asymmetric envelope is designed to produce, because τ_{release}≈1.5 s keeps the damping boost still partly engaged when the second impulse arrives.

5.7. Local Stability Certificate at the Standing Fixed Point

ACC's gates make it a switched system, which is why Section 2.1 places it alongside the switched-systems and hysteresis-switching literature [25, 26]. Everything reported so far is behavioral, in that the platform does not fall for 30 min in Section 5.1.4 and does not fall under any disturbance of Section 5.2, which is evidence rather than a stability statement. This section supplies one by locating the switching, characterizing the one live switch and examining the linearized closed loop.

At quiet stance the switching is inactive by a measured margin. Over the last 20 s of a 60 s settle, 19 of the 21 gates the controller exports sit at exactly 0 or exactly 1 at every control step, a bit-exact rather than an approximate saturation, while the two anchor gates that are not exported are reconstructed from controller state. The proximity gate's argument $|e_{\text{sag}}|$ peaks at 24.8 mm against its 50 mm knee, ranging over 16.1 – 33.2 mm across the five height variants for a margin of 1.5–3.1×, and the envelope gate's argument peaks at 1.5×10^{-7} m/s against its 0.25 m/s knee, a margin of 1.7×10^{6}. Two gates do sit inside their bands, the anti-twist gate at ≈0.53 and the centering gate at ≈0.986, but a smoothstep is C^{∞} in the interior of its band, so those are ordinary smooth feedback rather than switching. In a neighborhood of the fixed point every branch selection in the controller is therefore frozen, the closed loop is a single smooth map, and Lyapunov's indirect method applies. Chattering is not avoided here because the gates are C^1, it is avoided because the operating point stands far from every switching surface, and the margins above are how far.

The one switch that does fire is a contraction on both branches. The asymmetric envelope of Eq. (15) selects $\alpha \in \{0.35,\ 0.0067\}$ on the sign of $|v_{\text{sag}}| - \text{EMA}_{t-1}$, and in quiet stance that sign flips at essentially every step, which is the composition flagged in Section 4.1 as only C^0 in time. For a fixed input level the tracking error $e_t = \text{EMA}_t - |v_{\text{sag}}|$ obeys $e_{t+1} = (1-\alpha_t)e_t$, so $|e_{t+1}| \leq 0.9933\,|e_t|$ under either branch, making $V(e) = e^2$ a common Lyapunov function for the switched pair and the envelope globally exponentially stable under arbitrary switching at a worst-case rate set by the slow branch at τ=1.49 s. The classical obstacle, that switching among individually stable subsystems can destabilize the whole [25], cannot arise, and no dwell-time or hysteresis condition is required [26]. The C^0 remark is thus discharged twice over, since the discontinuity lives in a provably contracting scalar subsystem and reaches the loop only through a gate that is flat at unity across the entire quiet band.

The linearized closed loop is obtained by central-differencing the whole 100 Hz control step, meaning the control law together with its five physics substeps, about the settled state over an augmented vector of 57 coordinates. Of these, 30 are plant coordinates, namely 16 velocities and 14 configuration tangents once the two cyclic wheel angles are removed as in Section 5.4.2, 3 carry estimator memory and 24 carry controller memory. The adaptive-bias trim is held at its converged value because it is an 800-tap moving average and therefore memory rather than dynamics, and freezing it makes the one-step map time-invariant instead of 300-periodic. Table 6 reports that at all five heights the spectrum has exactly two eigenvalues on the unit circle, none outside it, and a reduced spectral radius of 0.997 87 – 0.998 18, corresponding to a slowest stable mode

Table 6
Local stability certificate at the standing fixed point, across the five calibrated height variants. Central-difference linearization of the 100 Hz closed-loop step after a 60 s settle; "Flat" counts exported gates that are bit-exactly saturated over the last 20 s. ρ_{red} excludes the two centre directions (lateral translation and deadbanded yaw), which are bounded separately by 300 s nonlinear probes.

Height variant	CoM z (m)	Gates flat	Centre dim.	Unst.	ρ_{red}	τ_{slow} (s)
Nominal	0.4069	19/21	2	0	0.99818	5.49
High small	0.4164	18/21	2	0	0.99787	4.70
High tiny	0.4128	19/21	2	0	0.99817	5.46
Low small	0.3973	19/21	2	0	0.99813	5.34
Low tiny	0.4009	19/21	2	0	0.99815	5.41

Produced by `scripts/acc_stability_certificate.py`; full spectra, gate traces and probe trajectories in `outputs/paper_verification/acc_stability_certificate.json`. Spectra are insensitive to the differencing step at the 10^{-7} level.

of 4.7 – 5.5 s. The classification is not a threshold judgement, because the gap from the unit circle down to the next mode is $1.8{\times}10^{-3}$, four orders of magnitude above the 10^{-7} sensitivity of the spectrum to the differencing step.

Both centre directions are identified and both are bounded by measurement. The first is lateral translation, which arises because a differential-drive base produces no lateral force to first order and appears at $|\lambda|{=}1$ to 10^{-12}, while the second is body yaw and is deliberate, since the heading channel carries an explicit deadband $\mathrm{ss}(|e_{\text{yaw}}|; 0.02\,\text{rad}, 0.08\,\text{rad})$ measured at exactly 0 here, so absolute yaw enters no control law while the error stays inside $\pm 1.15°$. Neither is left to the linearization, since perturbing each coordinate and running 300 s of the full nonlinear simulator returns a 1.000 mm lateral offset to 0.997 mm with no growth, while a 0.500° yaw offset creeps to 0.565° and would need 3.3 h to reach the deadband edge at which heading control re-engages. The lateral mode is the same one the 30 min run of Section 5.1.4 measures as 0.14 mm/hour of creep, so the certificate and the long run measure the same thing from two directions.

The linearization is finally checked against the plant it was taken from, random plant perturbations at 10^{-4}, 10^{-3} and 10^{-2} decaying in the full nonlinear simulator with nothing frozen at ρ between 0.995 13 and 0.995 47 against the 0.995 30 predicted for the anchor-integral mode at $\tau{=}2.12$ s, which is the mode a generic perturbation excites, so the agreement holds to 2×10^{-4} over two decades of amplitude. What this does not establish is equally definite, in that the certificate is local, rests on the nominal model at zero delay and clean sensing, estimates no region of attraction, and excludes the push results of Section 5.2 by construction, since a push is precisely what drives the gates off their flat regions and that is what the gates are for. What it does establish is that the point ACC parks at is exponentially attracting in 55 of 57 directions and neutral in the other two, with those two named, mechanistically explained and separately bounded.

5.8. Supplementary Video

Thirty-one clips accompany this paper under `paper/videos/`, and they are renders of the evaluation itself rather than a separate demonstration, since `scripts/render_paper_videos.py` drives the same harness that produced the corresponding table at the same seeds, so a clip and the number it illustrates cannot drift apart. The inventory covers quiet stance, ten push trials spanning the eight bearings of Fig. 4 together with one deliberate fall at 85 N, six drops from 10 – 100 cm, three curb straddles, eight ledge courses and three commanded height transitions. Each clip is one trial at one seed and therefore illustrates a regime rather than establishing a rate, the rates being the $N{=}10$ and $N{=}30$ columns of the tables cited above, while the noise and delay sweep, the classical-baseline comparison, the factorial ablation and the parking-offset model are not filmed at all because they distinguish their arms by numbers a viewer cannot see on a robot standing still. The per-clip protocol and the measured outcome are given in `paper/videos/README.md`.

6. Discussion and Conclusion

ACC achieves all demonstrated behaviors without neural network training and is hypothesized to carry a reduced sim-to-real transfer gap relative to learned policies, pending hardware validation [52]. Each gate was added after tracing a measured failure mode to its root cause, those modes being the bistable envelope, global-integral windup, parametric pumping, hard-leash oscillation and self-referential stiffness, and all thresholds were frozen prior to the ablation sweep of Table 3.

6.1. Statistical Power

All comparisons use Welch's t-test at a Bonferroni-adjusted $\alpha{=}0.0056$ over nine comparisons. The push ablation is run at $N{=}30$, where L0→L1 and L2.5→L3 are significant at $p{<}1 \times 10^{-15}$ with Cohen's d of 4.3 and 3.6, and the four subtractive removals are bounded rather than merely non-significant, since $|d|{\le}0.49$ and the shift on F_{min} stays below 1.4 N. The primary idle claims are stated as like-for-like ratios in Section 5.1, namely 59.4× sagittal, 22.6× planar and 5.0× lateral repeatability, all of them L0 against ACC at $N{=}10$ under one protocol, with the sagittal figure carried as the headline because that is the actuated axis. The S5 collapse is unambiguous, at 0/10 survival when idle and 10/10 at every push magnitude, and the flight and terrain rows are $N{=}10$ with a Clopper–Pearson 95% interval of [0.69, 1.00].

6.2. Limitations

The first limitation is that every result is obtained in simulation. All numbers are MuJoCo, hardware validation remains future work, and the sim-to-real gap is hypothesized to be reduced relative to learned policies rather than measured.

The second limitation, and the binding one for deployment, is timing. ACC as tuned requires a 100 Hz loop and an actuator latency clearing the 6 – 8 ms push-margin cliff of Section 5.6, it stops standing by 14 – 16 ms, and the 6.6 – 8 ms end-to-end budget estimated below sits inside the cliff bracket, so a non-real-time build is not guaranteed the nominal push envelope. Rate-independence is not claimed either, because at 50 Hz with coefficients recomputed for the longer step, giving $\alpha_a \approx 0.58$ and $\alpha_r \approx 0.013$, the platform does not stand at all and settles at a mean CoM height of 0.124 m against the 0.404 m nominal. Whether that is excitation of plant modes the faster loop suppresses or phase lag from discretizing the asymmetric envelope at Δt=20 ms is not resolved here and would require a loop-gain and phase-margin analysis. Both thresholds are measured on clean sensing, and the conjunction of noise with delay is bounded only by the 10 ms and 30 ms columns of the noise–delay grid.

The third limitation is that two of the calibrated envelopes are non-uniform. Oblique ledge descent has two survivable regimes separated by a dead band, as the 192-trial grid of Section 5.3 establishes, and since the best cell reaches only 75 % a clean launch is a favourable initial condition rather than a guarantee, while the 45° course at 32 cm is a reproducible 0/8 that the two-regime account does not predict. Upward height commands under simultaneous push are likewise restricted, since a +10 cm rise costs 69 % of $F_{\max}$ while descent is free at every amplitude tested. A control experiment attributes 35 % of that deficit to posture-map parameterization and the remainder to the physical cost of standing on a more extended leg, and the re-parameterization is not shipped because changing the posture map would require re-qualifying the standing suite against which it was frozen.

The fourth limitation is that the learning comparison is preliminary and one classical family remains untested. The model-free PPO run of 5.4M steps used one seed with no domain randomization [6, 7] and no hyperparameter search, which makes it a difficulty probe whose 85 % fall rate bounds nothing about RL on this platform. The comparison this controller is built for is residual learning over the prior rather than against it, and that comparison is not run here, as Section 1 states. Predictive control is the remaining untested classical family, but which part of it a receding horizon would repair can be identified rather than assumed. The full-state design of Section 5.4.2 fails in three separable ways, since it drifts, it holds one commanded height at a time, and it has almost no push envelope, and only the last two are open to a predictive formulation. A convex MPC over that same linearization, with terminal cost set to the Riccati solution of that design, emits exactly the LQR input at every step where no constraint is active [53], and across the entire pre-fall drift phase none is active, because the configurations that fall at 22 – 57 s run at 2.0 – 3.2 Nm torque RMS against a tightest actuator bound of 30 Nm, some 7 – 11 % of it, and under 2 % of the 150 Nm hip-pitch and knee bounds. The drift is therefore not a horizon artifact but a static offset that state feedback without integral action cannot reject, and the predictive formulations that do reject it, meaning offset-free MPC carrying an augmented disturbance state, reject it by adding integral action, which is what the anchor is. What a receding horizon would genuinely buy lies elsewhere, first in the push regime, where the torque bounds and the ±0.7 rad hip-roll limit do bind and constraint-aware preview has no counterpart among ACC's gates, and second across the height band, through relinearization at each step rather than about a single equilibrium. Those two are the comparisons left open, at a cost this paper can price, namely a dense QP over 30 states and 10 inputs at 100 Hz against the branchless 27 664-flop control step priced below. The task-priority QP whole-body controller of Section 5.5 does not resolve the precision and recovery trade-off, yet roll RMS is the single axis on which the blend beats ACC by 14 %, so the centralized formulation has merit that torque-level blending cannot harvest. Push ablations at N=30 resolve about 1 N, set by between-episode dispersion rather than by the search bisection, and no claim in this paper rests on a smaller difference.

The fifth limitation is that generality across morphologies is undemonstrated. Every result is for one 10-DOF platform with one set of gains, and the porting recipe of Section 5.6 is a hypothesis derived from this platform's sensitivity structure rather than a validated procedure.

6.3. Deployment Risks and Mitigations

ACC uses raw torque under a 400 Nm/s rate limit with standard sensing from an IMU and joint encoders. The binding latency figure is a performance cliff between 6 ms and 8 ms, across which $F_{\max}$ falls from 95.5 N to 51 N as Section 5.6 shows, whereas below 6 ms the cost stays under 2.4 %. The companion stability threshold is roughly twice as far out, since on clean sensing ACC holds station through 14 ms and falls by 16 ms as reported in Section 5.6. That ordering is the one the architecture predicts, because ACC's gates depend on physical state rather than on timing precision, the proximity gate being spatial at 15 cm, the asymmetric envelope being temporal but slow at $\tau_r \approx 1.5$ s, and the damping boost being state-driven rather than schedule-driven. Twice is nevertheless a factor of two rather than an order of magnitude, so a build that misses the latency target does not merely lose push margin gracefully, and commissioning should be done against the 6 ms cliff rather than against the 14 ms fall threshold.

The estimated worst-case budget for a non-RTOS build allows about 1 ms for sensor readout, about 2 ms for state estimation, at most 1.5 ms for ACC computation and about 3.5 ms for actuation latency, which totals roughly 8 ms end to end. Three of those four terms are assumed component figures while the ACC term is measured, on the same footing as the QP solve time reported for the discarded WBC baseline of Section 5.5. One control step of the shipped profile costs 1.11 – 1.22 ms mean and 1.27 – 1.52 ms p99 across five independent repeats of 5000 consecutive steps taken at the quiet-stance operating point, of which the compiled torque assembly itself accounts for 0.05 ms mean and the remaining 1.15 ms is the CoM and support-polygon computation together with Python-side dispatch. The 1.5 ms carried into the budget is the rounded upper end of that p99 range and it is doubly conservative, because the CoM estimate inside it is part of what the separate 2 ms state-estimation line already accounts for and is therefore charged twice.[15]

Wall-clock on a development host is nevertheless the wrong quantity to extrapolate to an embedded target, and the reason is visible inside the measurement, since five identical repeats give means spanning 0.11 ms, p99 values spanning 0.25 ms, and one worst single step of 3.6 ms caused by operating-system preemption rather than by anything the controller did. What is reproducible instead is the operation count. XLA's cost model reports 27 664 floating-point operations and 159 transcendentals per control step over a working set of 1001 scalars, identical to the last digit on every repeat. The step contains no iterative solve, no matrix factorization and no data-dependent control flow, every gate of Section 4.3 being a branchless select, so its worst-case execution time equals its average, which is the property a hard real-time loop requires and which the QP-based alternative cannot offer at all. At that count a 480 MHz Cortex-M7 with a single-precision FPU retires the step in about 0.07 ms when charged one cycle per operation and a conservative 40 cycles per transcendental, and a 100 MHz Cortex-M4 takes 0.34 ms. Compute reaches the 1.5 ms budgeted above only on a part slower than about 23 MHz, which is not a processor anyone would put a 100 Hz balance loop on. The compute term is therefore the one entry in the budget that gets smaller on the deployment target rather than larger, since roughly 95 % of the measured 1.2 ms sits outside the compiled kernel, in the CoM and support-polygon computation and the Python-side dispatch around it, work a C port either deletes outright or reduces to a few hundred operations of forward kinematics. Charging that line at an op-count-derived 0.1 ms instead of the host figure puts the end-to-end estimate at 6.6 ms. Both ends are reported, 8 ms instrumented and 6.6 ms projected, because the 4.5 ms of sensor readout and actuation between them is assumed in either case and is the larger uncertainty.

Read against the cliff at 6 – 8 ms, the instrumented 8 ms sits at the top of the bracket and the projected 6.6 ms at its bottom. Neither clears it outright, and what decides the matter is not compute but the 4.5 ms of sensor readout and actuation latency, which are bus and actuator properties rather than software. Two independent mitigations each move the loop to the clearing side. On a real-time operating system with a dedicated balance-loop microcontroller, whether 1 kHz RT Linux or bare metal, jitter is typically below 0.1 ms and 2 – 4 ms of budget is recoverable, which puts the loop at 2.6 – 4.6 ms on the projected figure and 4 – 6 ms on the instrumented one, below the cliff at either end. Separately, and without touching the schedule, doubling the scheduled sagittal damping $s_v k_{\mathrm{vel}}$ moves the knee out to 8 – 10 ms, which the projected figure clears outright and the instrumented one reaches, at negligible cost to undelayed quiet stance. The two act on different quantities, one shortening the loop and the other moving the cliff, so they compose rather than substitute. Real-time scheduling remains the recommendation and the loop-latency target is 6 ms end to end rather than 10 ms. The damping variant moves the push cliff only, leaving the 14 – 16 ms fall threshold unchanged, so it widens the performance margin without widening the safety one, and it is reported without being adopted for the reasons given in Section 5.6, since re-qualifying it over the 48-scenario suite is the work a hardware build would justify. Neither scenario has been validated on hardware, and latency characterization with an oscilloscope and a jitter histogram is required before deployment, being the measurement that decides which side of the cliff a given build lands on. The discarded WBC-assist variant of Section 5.5 is excluded from the hardware path on control grounds rather than compute grounds, because its QP solve costs 0.169 ms mean and 0.30 ms p99 over 481 000 solves and would fit the budget, yet admitting its torque degrades pitch by 2.5× and planar drift by 2.0×, while gating it to the point of harmlessness also gates it to 2.5 % of its permitted authority. On hardware the 0.294 mm sagittal repeatability is projected to degrade to 1 – 3 mm once encoder quantization, backlash, stiction, tire compliance and IMU noise are present, which is the same ordering the noise columns of the noise–delay grid display, and commissioning should proceed in stages from static torque to standing, then push recovery, then tethered flight and finally terrain.

6.4. Conclusion

We presented ACC, a two-channel torque assembly in which a PD balance core is augmented by a proximity-gated anchor integral, achieving 0.294±0.015 mm idle repeatability on the sagittal actuated axis, a 59× improvement over the proportional-only baseline under one unified protocol at $N=10$ in Table 2, together with omnidirectional push survival at $F_{\mathrm{min}}=82$ N and $F_{\mathrm{med}}=108$ N. Absolute accuracy on that axis is reported separately and fully accounted for, since the robot parks 27.0 mm from the commanded home and holds that standoff to 0.009 mm. The closed-form model of Appendix C predicts the offset as the static price of a deliberately bounded integrator memory, matching the measured torque balance to 0.45 %, and both available corrections are priced against the evaluation suite.

Six classical baselines, comprising four 4-state TWIP variants and two 6-state coupled variants, all fail without exception, so adding integral action to LQR-optimal feedback does not prevent the fall. Two control experiments test that comparison rather than assume it. Sweeping the coupled model's one empirical constant over an 80× range moves the outcome not at all, because the roll channel is saturated throughout, and a full-state LQR derived from the plant itself, with a sound derivation, a 0.54 % equilibrium residual, a stabilizable pair and a control weight swept over six decades, survives two orders of magnitude longer than the reduced-order designs and still falls, holding one commanded height at a time and recovering at most 20 N of push.

[15] Produced by `scripts/measure_control_step_cost.py`; raw output `outputs/paper_verification/control_step_cost.json`, which is the last of the five repeats. Host: Apple silicon (arm64), macOS, JAX CPU backend, Python 3.10.

ACC tolerates moderate noise, is insensitive to the actuator slew bound over an 8× span from 100 to 800 Nm/s with a 2 % spread in $F_{\max}$ and no resolvable change in lateral idle precision, and handles contact loss and terrain without neural network training. Its sharpest deployment constraint is actuator latency, since push margin collapses across a cliff between 6 ms and 8 ms that the 6.6 – 8 ms budget estimated for a non-real-time implementation lands inside rather than clear of, so real-time scheduling, a delay-hardened retune, or both are what put a build on the right side of it.

The contribution is thus a structured classical prior, an interpretable controller whose gating encodes the plant's measured failure modes, offered as such rather than as a competitor to a fully resourced learned policy. Everything claimed for it is a directly measured property of the controller standing alone, including the mechanism decomposition that assigns the precision to a specific term, the chirality of the push envelope, the closed-form account of the parking offset and the local stability certificate of Section 5.7, so the results do not wait on the residual stack the prior was built to serve. That stack is one downstream use among the future work, alongside hardware validation, EMA settle-time characterization, a properly resourced learning comparison with multiple seeds, domain randomization and a tuned reward, and extending conditional activation to locomotion.

Declaration of competing interest

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Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used Anthropic Claude in order to edit language and typeset the manuscript. All controller design decisions, experimental protocols, numerical results, and scientific claims are the authors' own. After using this tool the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

Data availability

The controller, the MuJoCo model, every evaluation script referenced above, and the raw result files backing each table are publicly available at `https://github.com/Thuong180702/Wheeled-bipedal-robot-simulation` at release tag `v1.0.0`, archived at `https://doi.org/10.5281/zenodo.21861318`. Each main results table names its producing script and its raw result file in the note beneath it, and the remaining sweeps are cited inline where they are discussed. The raw per-step records behind the whole-body-control results of Section 5.5 are collected under `outputs/wbc_postfix_evidence/` and those behind Appendix C under `outputs/parking_offset_evidence/`, each directory's `README.md` mapping every file to the table it produces and the latter also giving the worktree recipe for reproducing a gain arm without disturbing the shipped profile. The 31 supplementary clips of Section 5.8 are under `paper/videos/`, regenerated in full by `scripts/render_paper_videos.py`; that directory's `README.md` gives the per-clip protocol and the measured outcome each one illustrates.

A. Why Saturation-Triggered Anti-Windup Is Insufficient

Saturation-triggered anti-windup fires on actuator saturation, and on this platform the position error grows to 10 – 30 cm during a push while the wheels stay well within their 20 rad/s range, so the trigger never activates. A static deadzone either over-restricts or under-restricts, whereas ACC's proximity gate avoids both through a continuous smoothstep. State-gated conditional integration is itself standard practice in process control, but ACC couples spatial gating with asymmetric envelope-detected damping and so provides a temporal gating that static-threshold schemes lack. The two anti-windup augmented baselines are evaluated in Table 4, and a broader comparison against the full anti-windup taxonomy is deferred.

B. WBC Baseline Implementation Details

B.1. Provenance and Scope of the Claims

Every WBC number in Section 5.5 comes from a single campaign of the verified implementation covering 225 scenarios and 481 000 control steps, whose full metric set is Table 5, the degraded-solver control experiment contributing only the 85.85 % QP success rate and the 0.52 s time-to-fall taken from that build's comparison report in `docs/validation/V3_vs_V3_Assist_comparison_report.md`. Exactly one failure mechanism is asserted and it is verified directly in the source, namely that the posture task is a velocity damper rather than a position tracker, while two further mechanisms are excluded by the implementation itself. The QP is not linearized at a fixed operating height, since the mass matrix, the bias forces, the CoM Jacobian and the torso angular-velocity Jacobian are evaluated at the current state on every control step and the module's only linearization is the standard height-independent friction-cone pyramid, and the wheels are not bare point contacts, since lateral no-slip and forward rolling constraints are implemented and ran throughout the campaign. The architectural claim, that a fixed-task-priority formulation carries no regime gating, therefore rests on the invariance of the outcome to solver quality and task weighting rather than on any deficiency specific to this implementation.

B.2. WBC Task-Weight Ablation

The standalone WBC's failure could in principle depend on the particular task weighting adopted, so the WBC-only arm was re-run at four weightings over an identical scenario set of `random_push` at nominal height with seeds 201–205. The balanced default uses w_{torso}=3, w_{com}=3 and w_{posture}=1.5, and each priority mode raises its named weight to 10. No weighting rescues the baseline. The fallen fraction spans 97.3 – 97.4 % and upright time spans 0.56 – 0.59 s, a spread of 0.1 percentage points and 0.03 s, against ACC's 0 % fallen over the same scenarios. Roll RMS runs 92.86 – 95.28° and pitch RMS 1.52 – 2.51°, against ACC's 3.71° and 0.52°. Torso priority gives the best pitch and simultaneously the worst roll, because torso attitude is a single SO(3) task carrying equal roll and pitch gains, so raising its weight redistributes error between the two axes instead of reducing both. The four modes are `balanced_default`, `torso_priority`, `posture_priority` and `com_priority` in `wheeled_biped/wbc/offline_task_stack.py`.

B.3. Assist Gate Decomposition

The gated assist is inert because its authority gate is shut almost everywhere on this scenario matrix. The height factor $g_{\text{height}} = \exp[-((h - h_0)/\sigma_h)^2]$ uses h_0=0.53 m and σ_h=0.025 m and is hard-zeroed below a 1×10^{-3} cutoff, and four of the five height variants command a base-z far enough from h_0 that the commanded-height term alone drives the gate shut before the measured height is even considered, the factor running from 2.3×10^{-34} at `high_small` through 2.4×10^{-14}, 9.9×10^{-11} and 1.1×10^{-7} at `low_tiny`. Only `low_small` at a commanded 0.550 m leaves it open, at 0.53. Measured over 10 775 control steps on that one variant, the four state-dependent factors are 0.527 for height, 0.876 for stability, 0.998 for push and 0.999 for divergence, whose product is 0.461, while the per-joint agreement term A_j and role weight $K_{\text{role},j}$ reduce the admitted authority by a further 19× to $\bar{\alpha}$=0.0061 with a maximum of 0.0146, which is 2.5 % of $\alpha_{\max}$=0.25. The remaining shortfall between the factor product and the admitted authority is that the WBC's torque opposes ACC's on most joints and disagreement attenuates the blend. Both h_0 and σ_h are hand-tuned constants of ACC's own gate, σ_h having already been widened from 0.015 m, so they bound the authority ACC grants the WBC and are not a property of the WBC formulation, which is state-dependent and carries no fixed validity band.

B.4. Baseline Implementation Verification

A baseline reported to fail must be shown to be correctly implemented, or its failure becomes evidence about the code rather than about the formulation. Four properties on which a task-priority QP-WBC depends were therefore checked against executable references before the campaign of Section 5.5 was run, and all four pass. The analytic contact-point translational Jacobian matches MuJoCo's `mj_jac` element-wise at both wheel contact points to a maximum absolute error of 4.2×10^{-8} over the full matrix and 3.4×10^{-8} over the actuated block, against an acceptance bound of 1×10^{-3}, and the ipsilateral leg columns are nonzero with norms 0.042, 0.433, 0.403, 0.306 and 0.060 while the contralateral columns are exactly zero as the kinematics require, so there is no zero-gradient degeneracy. The cached QP cost builder is parameterized by `task_mode`, so the four weightings above are genuinely distinct problems, the one call site passing a literal being the task-independent constraint-only fallback that carries no cost terms by construction. The batch runner reads the ACC height reference from the scenario's `target_com_z` and no base-z path into the height command exists in the source, which matters because substituting one frame for the other would inject a 13.2 cm reference error under the convention of Section 3.1. Finally the adaptive-assist g_{height} gate is passed its own base-z target rather than reusing the CoM-z reference, so gate and controller agree on the frame, and the gate is nonetheless nearly shut on this scenario matrix for the tuning reason quantified in Section B.3. The checks are executable through `scripts/verify_wbc_audit_fixes.py` with archived output `outputs/wbc_audit_fix_verification.json`.

Because gate and controller agree on the height frame, the gate's near-total closure is a property of its tuning rather than of a frame mismatch, so the gated assist's numerical equivalence to ACC follows directly from a gate that is shut or nearly shut everywhere here and is not by itself evidence that the blend is harmless. The evidence about the blend is instead the ungated configuration of Section 5.5, which admits WBC torque on every step and degrades every stability metric but roll.

C. Static Torque Balance of the Sagittal Parking Offset

Section 5.1 reports a −27.0 mm sagittal parking offset on the axis the anchor integral regulates. This appendix derives the offset in closed form from a measured torque decomposition, validates the derivation against a 40× gain sweep and a five-point height sweep, and reports the measured cost of the available corrections. All runs use the idle protocol of Table 2 with identical seeds, a 5 s settle and a 20 s window, and the controller's internal sagittal torque decomposition is logged alongside the gravity moment about the wheel axles by `scripts/diagnose_parking_offset.py`, with raw records under `outputs/parking_offset_evidence/`. Sagittal is world y on this platform, so the x columns of the raw records are lateral.

C.1. The offset is neither gravity-driven nor a saturated integral

Two natural explanations fail on direct measurement, as Fig. 7a shows for N=10 with values averaged over the measurement window.

Gravity moment about the wheel axle	0.0040 Nm
CoM lever arm about the axle	0.050 mm
Anchor integral I	0.198 Nm (10.0 % of cap)
Integral clamp I_{max}	2.0 Nm
Pitch-loop torque τ_{pitch}	−1.2587 Nm
Position-loop torque $\tau_{position}$	1.2631 Nm
Adaptive bias trim τ_{trim}	0.0655 Nm
Controller position error e	−24.99 mm
Base parking offset $\bar{s}$	−26.96 mm

Gravity is not the load. The moment is 0.0040 Nm because the CoM sits 0.050 mm from the axle line, which is not a coincidence of tuning, since a wheeled inverted pendulum that has stopped accelerating has necessarily brought its contact patch under its CoM whatever its absolute position. The parking offset is a displacement of the whole machine relative to the commanded home rather than a static lean, so no relation of the form $\tau_{gravity}(\bar{s}) \approx \tau_{integral}$ can hold.

The integral is not saturated either. I reaches 10.0 % of its clamp during quiet stance and never approaches it, so the offset is neither a windup nor an anti-windup artifact and cannot be relieved by raising the clamp. This is consistent with the ablation result of Table 2, where removing the proximity gate entirely in S4 reproduces ACC bit-for-bit. What does balance is τ_{pitch} against $\tau_{position}$, to within 0.005 Nm.

C.2. Origin of the constant pitch torque

The sagittal pitch term is $\tau_{pitch} = k_{p,\theta}(\theta - \theta_{ref})$ with $k_{p,\theta} = 50\,\mathrm{Nm/rad}$ at idle, and decomposing the commanded reference at the nominal standing height h=0.4040 m gives the following.

Feedforward equilibrium-pitch calibration	+3.4963°
Support-position outer loop (PD, k_p=0.8145 °/m)	−0.0203°
Low-band support term (gated off at this height)	0.0000°
Commanded reference θ_{ref}	+3.4760°
Measured body pitch θ	+2.0336°
Residual $\theta - \theta_{ref}$	−1.4424°
$\Rightarrow \tau_{pitch}$	−1.2587 Nm

The decomposition closes to four decimal places, so the offset originates in a 1.44° calibration error of the feedforward equilibrium-pitch map at this operating point rather than in the anchor at all. The outer loop does observe the 25 mm error yet contributes only 0.02° against the 1.44° it would need, because its integral channel is disabled and it therefore has finite DC gain as well.

C.3. Closed form

The sagittal wheel torque during quiet stance is $\tau_{pitch} + \tau_{position}$ with the position law unsaturated, since $|\tau_{position}| \approx$ 1.3 Nm against a 4.0 Nm cap, so that

$$\tau_{position} = -k_{pos}\, e + I + \tau_{trim}. \tag{25}$$

The implemented anchor integral is leaky. Eq. (8) writes the ideal integral for clarity, whereas the discrete update applied at every step is

$$I_{k+1} = (1-\lambda)\big(I_k - k_i\,\Delta t\, g_{prox} g_{env}\, e_k\big), \tag{26}$$

with λ=5 × 10^{-3} per step as listed in Table 1. A leaky integrator is a first-order lag rather than an integrator, so at a fixed point with the gates open Eq. (26) gives $I^* = -k_{I,dc}\, e$ with

$$k_{I,dc} = k_i\,\Delta t\,\frac{1-\lambda}{\lambda} = 4.0 \cdot 0.01 \cdot \frac{0.995}{0.005} = 7.96\,\mathrm{Nm/m}, \tag{27}$$

which is one fifth of k_{pos}=40 Nm/m rather than infinity. Substituting into Eq. (25) and imposing zero net sagittal torque yields

$$\begin{aligned} e_{ss} &= \frac{|\tau_{pitch}| - \tau_{trim}}{k_{pos} + k_i\,\Delta t\,(1-\lambda)/\lambda} \\ &= \frac{1.2587 - 0.0655}{40 + 7.96} = 24.88\,\mathrm{mm}, \end{aligned} \tag{28}$$

against 24.99 mm measured, an error of 0.45 %. The remaining 2.0 mm between e and the base offset $\bar{s}$ is accounted for by the base-frame versus support-centre convention gap, so the two figures are consistent once expressed in the same frame.

The conclusion is structural rather than incidental, in that no element of ACC's sagittal position chain has infinite DC gain. The anchor integral is leaky by design and the outer loop's integral channel is disabled, so a constant disturbance torque must be paid for with a constant position error, and the 27 mm is the price of that design choice expressed in millimetres per newton-metre.

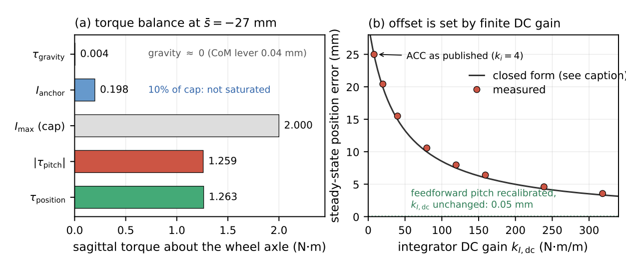

Figure 7: Static torque balance of the sagittal parking offset. (a) Measured sagittal torques at the $\bar{s}$=−27 mm parking point. The gravity moment about the wheel axle is 0.0040 Nm and the anchor integral sits at 10 % of its clamp, so the offset is neither gravity-driven nor a saturation artifact; the balance is between the pitch and position loops. (b) Steady-state position error against integrator DC gain $k_{I,dc}$ of Eq. (27), swept over $k_i \in [4, 160]$ at fixed leak. The curve is Eq. (28) evaluated with the baseline τ_{pitch}; using each run's own measured τ_{pitch} the closed form tracks all twelve gain and height points to within 0.92 %. The dotted line is the same controller with the feedforward pitch reference recalibrated and $k_{I,dc}$ left unchanged.

C.4. Validation across a forty-fold gain sweep

Sweeping k_i from 4 to 160 at fixed leak moves $k_{I,dc}$ over 40×, and the offset follows Eq. (28) throughout as Fig. 7b shows.

k_i	$k_{I,\text{dc}}$ (Nm/m)	e meas. (mm)	e pred. (mm)	window jitter (mm)
4	**7.96**	**-24.99**	**-24.87**	**0.298**
10	19.90	-20.42	-20.34	0.302
20	39.80	-15.49	-15.54	0.147
40	79.60	-10.57	-10.67	0.208
60	119.40	-7.96	-8.01	0.168
80	159.20	-6.40	-6.41	0.094
120	238.80	-4.59	-4.58	0.114
160	318.40	-3.56	-3.56	0.066

Steadiness improves monotonically with k_i over this range rather than degrading, the opposite of the trade-off measured at k_i=0 in Table 2, which indicates that the 3× steadiness cost reported there is paid at the first newton-metre of integral authority rather than progressively. Every entry is measured on clean sensing with no actuator delay, which Section C.5 shows is exactly the condition under which raising k_i looks free.

C.5. Why the gain is not raised: the noise–delay cost

The sweep above and the push protocol below both make k_i=160 look free, since the offset falls by 79.5 %, clean-idle jitter improves, and neither F_{min} nor F_{med} moves. Those three probes share a blind spot, in that all are run on clean sensing and the push probe is a large transient rather than a sustained small one. Re-running the two cells of the noise–delay grid that combine medium sensor noise with actuator delay, at N=20 per cell and identical seeds across arms, prices the change.

k_i	$k_{I,\text{dc}}$ (Nm/m)	$\bar{s}$ (mm)	offset removed	falls (of 40)	Fisher vs. k_i=4
4	**7.96**	**-26.96**	—	**7**	—
20	39.80	-17.47	35.2 %	13	p=0.196
40	79.60	-12.55	53.5 %	10	p=0.586
80	159.20	-8.37	68.9 %	14	p=0.126
160	318.40	-5.53	79.5 %	17	p=0.027

A Cochran–Armitage trend test on $\log_2 k_i$ gives z=2.34 at p=0.019, so the fall rate rises with integral gain across the sweep, and pooling every arm above the shipped setting gives 54 falls in 160 against 7 in 40 at p=0.055. Because the cost is graded rather than a threshold, selecting the largest gain that is not individually significant would be an artifact of sample size, since k_i=40 carries p=0.59 only because 40 trials cannot separate 17.5 % from 25 %. Zero-delay cells stay clean at every gain and the idle, push, flight, terrain and rate-limit metrics are flat across the sweep, so the regression is specific to noise and delay acting together. One trap is worth naming, namely that the idle column of that grid averages survivors only, so in these two cells the higher-gain arms report better idle numbers while falling more often.

The mechanism follows from Eq. (26). The leaky integrator has transfer function $k_i\Delta t/(\lambda + s\Delta t)$, giving a DC gain of $k_i\Delta t/\lambda$ and a corner at $\lambda/\Delta t \approx 0.5\,\text{rad/s}$. Raising k_i scales the response at every frequency, so it lifts mid-band gain, carrying the 90° lag of an integrator, in precisely the band where the 10 ms actuator delay of those cells has already consumed the phase margin. Lowering λ raises DC gain alone and leaves mid-band untouched but costs the memory horizon instead, as the next subsection shows. Neither single parameter separates the two roles, and doing so would require a lead-compensated or explicitly bandwidth-limited integrator, which is a structural change to a controller whose present form is qualified over a 48-scenario suite and is therefore left as future work.

C.6. The leak sets memory, not accuracy

Because $k_{I,\text{dc}}$ depends on both k_i and λ, reducing the leak also raises DC gain, but it is the wrong knob. Cutting λ from 5×10^{-3} to 5×10^{-4} raises $k_{I,\text{dc}}$ to 79.96 Nm/m and does reduce the offset to −16.31 mm, yet window jitter degrades twelvefold from 0.298 mm to 3.463 mm and the measured error of −14.34 mm does not reach its DC prediction of −10.41 mm, because the forgetting time constant τ=$\Delta t/\lambda$ has grown to 20 s and no longer settles inside the measurement window. Since τ is independent of k_i the two parameters separate cleanly, k_i setting DC gain and λ the memory horizon, and the shipped pair is chosen on that basis, λ=5×10^{-3} fixing the 2 s forgetting time that keeps the integral inside its measurement window while k_i=4 is set not for clean-idle steadiness, which the sweep shows is flat in k_i, but for margin against the conjunction of noise and delay quantified in Section C.5. The accuracy consequence of that pair is the 27 mm offset, the price of the memory horizon rather than of the gain.

C.7. Two measured corrections and their cost

The height dependence of the calibration error was checked on the five posture-calibrated height variants by applying the correction measured at each point. The nominal row is N=10 under the full idle protocol while the four off-nominal heights are single trials, whose run-to-run spread the nominal row bounds at 0.011 mm SD.

h (m)	correction (°)	$\bar{s}$ before (mm)	$\bar{s}$ after (mm)	reduction
0.394	−1.622	−28.77	−1.16	96.0 %
0.399	−1.974	−34.53	−1.20	96.5 %
0.404	**−1.461**	**−26.96**	**−2.02**	**92.5 %**
0.409	−0.943	−19.29	−2.83	85.3 %
0.414	−1.295	−25.11	−2.92	88.4 %

The controller's own position error falls to ≤0.12 mm at every height and the anchor integral to 0.4 % of its clamp, so what remains is the 1 – 3 mm frame-convention gap. The required correction is not a constant, since it spans 0.94 – 1.97° over 2 cm, which makes this a per-height recalibration of the feedforward map rather than a single scalar.

Both corrections were then put through the full push protocol of Section 5.2, meaning N=10 repetitions at each of eight bearings with a binary search over 10 – 160 N at 5 N tolerance and Welch's t against the published arm.

Arm	F_{min} (N)	F_{med} (N)
ACC as published (k_i=4)	82.8(10)	116.8(47)
k_i=40	81.8(18) p=0.14	116.8(70) p=0.98
k_i=160	81.9(27) p=0.32	115.3(36) p=0.43
Feedforward pitch recalibrated	82.5(12) p=0.55	112.4(49) p=0.05

The published arm reproduces the weakest bearing of Section 5.2 at 82.8 N here against 82.1 N there, a difference below 1 N on independent seeds, which validates the harness. Raising k_i by 40× costs nothing measurable on either push statistic and removes 79.5 % of the offset, while the feedforward recalibration removes 92.5 % and leaves F_{min} unchanged but shows a 4.5 N reduction in F_{med}, or 3.8 %, at p=0.051, which at this sample size is reported as unresolved rather than as null.

Neither correction is promoted into the shipped profile, and in the k_i case the reason is not consistency of the evidence base but a measured regression, because the push protocol is blind to the failure mode that the gain change introduces and that mode appears only when sustained sensor noise and actuator delay act together, as Section C.5 establishes. This is the substantive result of the appendix. The 27 mm offset is the position error that a deliberately leaky and deliberately low-gain integrator pays in exchange for a 2 s forgetting horizon and for phase margin against delay. Both alternatives were measured across the full evaluation suite rather than on the accuracy metric alone, which is what exposed the price. The feedforward recalibration remains the more promising route since its cost is confined to one borderline statistic, and resolving that at a larger sample, together with testing whether a bandwidth-limited integrator can raise DC gain without raising mid-band gain, are the two concrete steps toward closing this limitation.

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